# Eclipses and the Memory of Revolutions: Evidence from China

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**Abstract**

We study the historical roots of anti-government protests in China. We document that Chinese counties with a higher incidence of peasant uprisings against local government officials during the Qing dynasty period (1644-1912) had a higher participation in the protests that led to the 1911 Revolution and the fall of the imperial rule. To generate exogenous variation in the incidence of past protests, we exploit differences in the visibility of solar eclipses across counties. In the Confucian tradition, solar eclipses were considered negative divine signals on the legitimacy of rulers, facilitating the coordination of anti-government actions.

**Keywords:** Social Protests, Persistence, Qing dynasty, Confucianism.
**JEL Classification:** D74, O53, N3

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# 1 Introduction

Throughout history, social protests have brought issues at the forefront of public debate, influenced policy-makers' agendas, and, in some instances, forced regime changes. Although many communities within a country often share similar grievances towards their government, protests do not erupt with the same intensity everywhere. In this paper, we bring evidence on a specific determinant of observed differences in the incidence of protests and the persistence of such differences over time: the existence of a historical tradition of anti-government actions within a given community.

We focus on the wave of protests that erupted across China in the first decade of the 20th century and that culminated with the 1911 Revolution and the consequent end of the Chinese Empire. The Qing dynasty had ruled China since 1644. Over nearly three centuries, it faced numerous episodes of local unrest, often in the form of uprisings by farmers against local government officials. We investigate whether a local tradition of social uprisings built over centuries of Qing rule affected the local participation in the wave of protests that culminated in the 1911 Revolution.

We construct a new dataset containing the date and location of peasant uprisings that occurred during the Qing dynasty period starting from county chronicles. County chronicles record major events that took place in a given county, including insurgency actions, and were compiled by independent, reputable local intellectuals. Our dataset includes 1,956 rebellions that occurred under the Qing dynasty between 1647 and 1911. Rebellions were frequent in rural China and played an important role in several major political changes in Chinese history. Poverty and hunger were the primary causes of peasant uprisings, which targeted local elites and government officials for their unrestrained exploitation of local farmers via over-taxation (Long, 2016; Wang, 1884).

The main identification challenge is that unobservable characteristics that make certain communities more rebellious in the past might also make them more rebellious decades, or even centuries later. We overcome this challenge by exploiting plausibly exogenous variation in the incidence of rebellions generated by the visibility and magnitude of solar eclipses across Chinese counties during the Qing dynasty period. In the Confucian tradition, solar eclipses were considered a negative divine signal on the legitimacy of rulers. We argue that eclipses acted as a salient and observable signal with a shared interpretation among the local population, facilitating the coordination of anti-government actions and thus increasing the probability of local protests in communities that observed them.

We merge data on the year and location of rebellions during the Qing dynasty period with geo-located historical data on the areas of visibility and magnitude of solar eclipses from the

US National Aeronautics and Space Administration (NASA). Solar eclipses are a plausibly exogenous source of variation because their magnitude, areas of visibility and timing are solely determined by the relative position of the sun, the earth, and the moon. In addition, different from natural disasters, solar eclipses have no direct destructive impact on physical capital or agricultural productivity. Consistently, we show that their timing is uncorrelated with weather or economic shocks.

In the first step of our empirical analysis, we test whether the visibility of solar eclipses affects the incidence of peasant uprisings. Using a panel specification that covers the Qing dynasty period, we find that counties in the totality zone of a solar eclipse – i.e. the relatively thin strip of land in which the view of the sun is completely blocked by the moon – are between 12 and 13 percentage points more likely to experience a rebellion in the eclipse year relative to counties outside of the totality zone but located within the same province. Eclipses affect the contemporaneous probability of uprisings, with no visible pre-existing trends. The effect is significantly larger in counties with a Confucian temple, which we use as a proxy for the local diffusion of Confucianism.

Next, we study the effects of a historical tradition of anti-government rebellions on participation in the wave of protests that led to the 1911 Revolution. We provide direct empirical evidence of such a connection by exploiting plausibly exogenous variation in the incidence of past rebellions generated by solar eclipses. Our findings indicate that past incidence of social unrest was indeed an important driver of participation in one of the key regime changes in China's history. The estimates indicate that counties with a standard deviation larger number of rebellions per 1,000 people during the Qing dynasty period (up to 1899) experienced 1.3 more protests per 1,000 people in the decade that led to the revolution (1902 to 1911).

In the last part of the paper, we test for persistence in the very long run using data on social protests in present-day China. Our main source is the Chinese Academy of Social Sciences (CASS) database, which records social protest events involving at least 100 participants at the county level between 2001 and 2013. We complement this with the China Labour Bulletin (CLB), which records labor disputes, the Armed Conflict Location and Event Database (ACLED) and the Global Database of Events, Language and Tone (GDELT), which record social protests from media reports. Although coefficients are noisier than those estimated for the 1911 Revolution, we find a broadly consistent picture across all data sources: regions with more eclipse-driven past rebellions tend to display higher protest activity in the first two decades of the 21st century.

We explore potential mechanisms behind the documented persistence. First, we investigate the role of collective memory. The literature has documented that cultural norms and

beliefs can persist within certain communities over long periods of time.[1] We collect new data on measurable signs of inter-generational transmission of the memory of anti-Qing rebels of the past. These include physical structures such as temples and memorials celebrating prominent individuals that fought the initial Qing invasion of China, and the presence of a positive recorded memory of these early anti-Qing leaders as captured by the description of their actions in local chronicles (what we hereafter refer to as legends).[2] We document that counties with more past rebellions are more likely to have temples or legends celebrating anti-Qing martyrs. The empirical literature has shown evidence that collective memories from the past can be re-activated by current events, even after being dormant for long periods of time.[3] This mechanism is consistent with models of choice such as Bordalo et al. (2020), where the memory of past experiences serves as an anchor for individual decisions and influences how they react to new information.

Second, we examine the governance and administrative response of the Qing state that might have affected future propensity to protest. The Qing government maintained a standardized county classification system (*chong, fan, pi, nan*) that rated each county on strategic location, administrative complexity, tax collection difficulties, and crime. Using eclipses in the early Qing period as instruments for early rebellions, we find that rebellion-prone counties were subsequently classified as more difficult to govern. This administrative labeling, which lasted throughout the Qing dynasty period, could have reinforced the conditions for future grievances.

Finally, we consider the role of social capital. Past rebellions may have strengthened local organizational capacity, by fostering clan networks or other forms of community solidarity that facilitated future collective action. We test this channel by examining whether eclipse-driven past rebellions predict the local density of clans, but find no evidence of this channel. We also find imprecise estimates on the effect of protests on several measures of economic development, which suggests that local economic conditions are unlikely to be the primary driver of persistence.

We present several robustness tests on the empirical results. Following the approach of Madestam et al. (2013), we conduct a placebo exercise using NASA's 5,000-year eclipse

[1] See Voth (2021) for a recent review. Voigtländer and Voth (2012) document that German towns with a higher incidence of pogroms against Jews in the Middle Ages displayed significantly more anti-Semitic attitudes in the interwar period of the 20th century. They argue that persistence might be partly explained by local anti-Semitic traditions and symbols that remained in the local culture for centuries.

[2] See the related work by Michalopoulos and Xue (2021), which studies the role of oral traditions (folklore) in explaining why certain social and cultural attitudes persist within a given community.

[3] For example, Ochsner and Roesel (2024) document that Austrian municipalities that were pillaged by Turkish troops between the $16^{th}$ and the 17th century were more responsive to a 2005 right-wing populist campaign against Turks and Muslims. On the role of memory in shaping local collective action see also the related work by Gauthier Marchais et al. (2021) and Dell and Querubin (2018).

catalog. We re-estimate the first stage using eclipse patterns from temporal windows of the same length as the Qing dynasty but shifted outside the actual sample period. We find that only eclipses from the Qing period predict rebellions that occurred during the Qing period, while eclipse patterns from other periods produce coefficients that are uniformly smaller and, in most cases, not statistically significant.

While the timing and path of solar eclipses are determined by celestial mechanics and thus plausibly exogenous, the astronomical literature has shown that the probability of experiencing a total solar eclipse is not perfectly uniform across locations on Earth. In particular, Meeus (1982) documents that eclipse frequency varies systematically with latitude, and Buus (2014) shows that eclipse duration tends to be longer closer to the equator. If other determinants of protest persistence are also correlated with latitude, this spatial variation could confound our estimates. To address this concern, we control for latitude and for the long-run frequency of eclipses in each county outside of the Qing dynasty period in all cross-sectional specifications. Our results are robust to the inclusion of these controls.

To account for potential spatial correlation in residuals beyond the prefecture level, we report Conley-adjusted standard errors (Conley, 2010) at distance cutoffs of 50, 100, 200, and 300 kilometers for all main results. Our findings remain robust across all distance thresholds.

*Related Literature*

Our paper is related to several streams of the literature. First, the mechanism linking solar eclipses to the probability of protests is related to the literature on coordination in protest actions and on how individuals' decision to participate in protests is influenced by the decision of others. Several papers postulate that the actions of a "pilot" of revolutionaries can act as a signal for other people to join. De Mesquita (2010) studies how a small number of rebels ("vanguard") can use violence as a form of public information, signaling to others that there are strong feelings of opposition to the government. Similarly, Lohmann (1994) exploits the events preceding the fall of the Berlin Wall to study how potentially costly (but peaceful) public demonstrations can result in an informational cascade, drawing additional participants as they observe fellow citizens demonstrating in increasing numbers. Cantoni et al. (2019) use a field experiment in Hong Kong to study how information about others' plans to protest affects an individual's decision to participate, and find evidence of strategic substitutability, namely that information of higher-than-expected planned participation by others has a negative effect on an individual's own participation. Our paper exploits the insight of this literature to rationalize the empirical link between solar eclipses and rebellions during the Qing dynasty period.[4]

[4]The literature on coordination and coordination failure is vast and applies to a wide range of economic

Our work is related to the literature on persistence in political engagement. Bursztyn et al. (2021) use a field experiment during the Hong Kong anti-authoritarian movement to show that individuals participating in the 2017 protests were significantly more likely to also participate in the following year's protests. This is an important question because the success of a political movement often depends on its ability to sustain political engagement over time. In other related work, Madestam et al. (2013) use weather shocks to instrument for attendance at tea party protests in 2009 and find that participation in such protests in 2009 affected votes for the Republican Party in the 2010 US midterm elections.[5] Overall, these studies focus on short-run persistence in protest participation. Our paper differs from this literature by focusing on the persistence of a culture of protest against government authority and by documenting that the effects of such culture can materialize over a long period of time. Indeed, our findings indicate that rural communities where peasants were more prone to challenge the local authority of the Qing were also more likely to participate in the revolution that led to the end of this imperial dynasty decades or even centuries later.

Our paper is also related to the large literature in history, sociology and economics studying the determinants of the fall of the Qing dynasty. Such determinants included a series of military defeats with the consequent burden of reparations in the early 20th century, an economy characterized by high inflation, a weakening of the central government capacity and a series of hurried reforms which generated broad discontent (see Esherick and Wei (2013) and Zheng (2018) for a discussion). One such reform was the elimination of the imperial examination to recruit civil servants, the traditional means of social mobility during over a millennium of imperial rule (Franke, 1960, 1970). Bai and Jia (2016) provide direct evidence that the abolition of the imperial civil service exam by the Qing court in 1905 fostered participation in subsequent social protests due to the increase in resentment of individuals that aspired to use the exam to become part of the imperial bureaucratic elite.[6] Scholars of Chinese history have argued that a legacy of peasant rebellions might have constituted fertile

phenomena, from bank runs (Diamond and Dybvig, 1983; Shiller et al., 1984; Hertzberg et al., 2011) to asset price fluctuations Azariadis (1981); Cass and Shell (1983); Obstfeld (1986); Tirole (1985); Grandmont (1985); Azariadis and Guesnerie (1986); Guesnerie (1986); Woodford (1986, 1990); Farmer and Guo (1994); Benhabib and Farmer (1999); Morris and Shin (2002). For evidence on the coordination mechanism from lab experiments see Mehta et al. (1994a,b); Duffy and Fisher (2005); Fehr et al. (2019); Arifovic and Jiang (2019). See Manacorda and Tesei (2020) and Enikolopov et al. (2020) for evidence on the role of social media and telecommunication technologies as coordination mechanisms facilitating protest participation. More generally, our work is related to the literature on the determinants of social unrest: see Gehlbach et al. (2016) for a recent review, and related work by Autor et al. (2020), Algan et al. (2017), Fetzer (2019), and Ponticelli and Voth (2020)

[5]See also Aidt and Franck (2015) and Collins and Margo (2007) for evidence on the effect of protest participation on voting behavior and economic outcomes.

[6]See, on this, also Hao et al. (2022), which argues how the abolition of the civil service exam led to a deterioration in the quality of local governance, and a consequent increase in protests against local elites.

ground for local participation in the revolution that ended the Qing dynasty (Perry, 1980).[7] We provide causal evidence on this relationship and quantify its impact.

Finally, our identification strategy is related to the literature using natural phenomena as a catalyst for change in religious and political power (Lipset, 1959; Barro, 1999; Acemoglu, 2006). Chaney (2013) analyzes the relationship between the flooding of the Nile and related shifts of political power in Egypt. He suggests that a religious figure's power increases during perilous economic circumstances brought by the Nile, and that this figure is less likely to be replaced, mainly due to their potential for coordinating a revolt. Belloc et al. (2016) find a similar result in the case of mild earthquakes in the south of Italy between 1000 and 1300, which led to an increase in the perceived power of politico-religious leaders to restore social order, delaying the transition to communal institutions. In the context of China, Bai (2023) shows that mild earthquakes generated higher political instability – as proxied by conflicts – between the XIV and XIX centuries.[8] In other related work, Sun and Li (2024) investigates the relationship between political legitimacy and economic development during over 2000 years of imperial China, using solar eclipses as plausibly exogenous shocks to political legitimacy. They document that, while eclipses had no direct effect on economic performance, they amplified the negative impact of weather shocks on the local economy when they occurred in the same year. While they study the contemporaneous interaction between eclipses and weather shocks on economic outcomes, we use eclipses as instruments for the historical incidence of rebellions and study the long-run persistence of protest behavior across generations.

The rest of the paper is organized as follows. Section 2 provides the historical background. In this section we discuss the interpretation of solar eclipses in Confucianism, and their potential as a coordination device for protest actions. We also discuss the type of rebellions and the motives behind them during the Qing dynasty period and the 1911 Revolution. Section 3 describes the data sources. Section 4 presents the empirical strategy and discusses the identification assumptions. Section 5 describes the main empirical results on protests during the Qing dynasty period and their effect on the 1911 Revolution, while Section 6 discusses and empirically investigates potential mechanisms. Finally, Section 7 presents suggestive evidence on the persistence of protest behavior in the very long run.

[7] See discussion in Perry (1980, p. 7), who notices that "[t]he notion of a positive connection between a long tradition of rebellion and the potential for modern revolution is a familiar theme in general theories of peasant revolt".

[8] Ticku et al. (2023) offer a complementary explanation for the relationship between negative weather shocks in India and recorded temple desecration, a preemptive measure by a Muslim ruler to avoid such coordinated revolutions within their Hindu constituencies.

# 2 Historical Background

## 2.1 Solar Eclipses as a Coordination Device

Previous literature has argued that random and otherwise irrelevant events ("sunspots") can act as a coordination device among agents as long as the signal is salient and observable by many agents (Shelling, 1960) and a common belief about the meaning of the signal exists among the agents (Duffy and Fisher, 2005). We argue that solar eclipses observed during the Qing dynasty period fit these two criteria: they are a salient signal from the perspective of observers on the earth and have a shared interpretation as the manifestation of the heavens' opinion about a ruler.

Solar eclipses occur when the paths of the sun and the moon intersect so that the sunlight is temporarily blocked by the moon, casting shadows on earth. A total eclipse, in which the sun is completely blocked, can only be observed in a certain area of the earth at any given time during an eclipse event. Referred to as the zone of totality, these areas are usually several thousand kilometers long and less than one hundred kilometers wide, with individuals inside the zone of totality observing the eclipse within the same day, several hours apart. During a total eclipse, the sun is blocked, day becomes night, and stars become visible for as long as one hour.

Confucianism was the dominant ideology in ancient China, including the Qing dynasty period, and continues to play a major role in shaping the minds and actions of Chinese society today. In the Confucian tradition, a solar eclipse is a signal regarding the legitimacy of the ruler. An important saint in Confucianism, Mencius, believed the legitimacy of emperors to come by endorsement from heaven, under the condition that the emperor takes care of his people.[9] Those rulers who failed to do so would be abandoned by heaven, leaving the people with the right to overthrow them.

According to Confucianism, other natural anomalies such as natural disasters and celestial phenomena are also indications of the heavens' opinion about the ruler. However, a solar eclipse represents the most serious condemnation from the heavens. As the sun is regarded as the sign of the emperor (Li, 1990), an eclipse of the sun can be related to his moral faults. This idea dates back to the earliest books compiled by Confucius himself during the Spring and Autumn period of China. For instance, "The state without good governance could bring

[9]Mencius proposed three intersections between Confucianism philosophy and politics: 1) That heaven is what grants an emperor's power (Mengzi 5A5), 2) That an emperor needs to exercise benevolent governance in accordance with the will of heaven (Mengzi 1B6, 2A5, 2B1), and 3) That an emperor who does not govern accordingly will be condemned by heaven, warning that righteous fury can come by way of natural anomaly. At those times, any consequent revolutions are legitimate (Mengzi 1B8, 2B9).

in the disaster of the solar eclipse" (Zuozhuan, Zhao, 7) or "The sun and moon did not follow common rules because the governance of the state was not good" (Shijing, Xiaoya, In October). During the Han and later dynasties, warnings of the eclipse were also common: "The most notable is Heaven and the ultimate change that could happen to the emperor is the solar eclipse" (Wuxing, Houhan, Ji, 2003, book 7, PP429).

The association between solar eclipses and ruler's illegitimacy was popularized in the era of Keju examinations. Keju examinations were the device used in imperial China to select civil servants, and the main, if not the only, channel of social mobility. The interpretation of solar eclipses appears in all nine works of the Confucian canon, which constitutes a large portion of the question pool at all levels of the Keju examinations. This means that more than 5% of the population, attracted by the economic and social benefits associated with passing the imperial examination, would spend a large share of their adult lives memorizing and understanding the Confucian canon and were therefore familiar with this interpretation of solar eclipses. The candidates, whether successful or not, usually enjoyed a high social status and had a strong voice in local affairs due to their ability to read and write. In this sense, when commoners had difficulty understanding what an eclipse meant, the eclipse-illegitimacy interpretation they heard from respectable local intellectuals was likely to be both intuitive and convincing to them.

Because of their shared interpretation, solar eclipses were feared by emperors. As a preemptive measure, it was an established tradition for emperors to issue "self-condemnation" scripts, asking for the pardon of the heavens whenever a solar eclipse was observed. The first self-condemnation script was issued by emperor Wen during the Han dynasty, who witnessed two eclipses during his reign. Emperor Wen declares: "I am incapable of ruling and taking care of my entire population and therefore bring about a solar eclipse that illustrates how large my sins are". Issuing self-condemnation scripts after the sight of a solar eclipse became a royal practice passed down to later generations of emperors. In addition to these scripts, emperors usually took measures to improve their administration, please their subjects and prevent local rebellions. Those measures included cutting local taxes, forgiving inmates of milder crimes, allowing people to bring their petitions to the capital, or increasing local quotas for the civil service exam (Zang, 2015).

Due to its political sensitivity, the study of astronomy was largely monopolized by the imperial court via the Bureau of Astronomy. Access to instruments and books was restricted to affiliated astronomers, who held lifetime and hereditary appointments. This might explain why the predictions of lunar and solar eclipses – one of the most important tasks the Bureau of Astronomy was in charge of – were still very inaccurate even in late Qing China (Lyu and Shi, 2003b,a). Although astronomers had a clear understanding of the natural causes of solar

eclipses, such events kept being considered an ominous event among the population. Due to the high levels of illiteracy and the absence of modern science education, the vast majority of Chinese people during the Qing dynasty period had scarce knowledge of astronomy.[10]

## 2.2 Social Protests during the Qing dynasty period

Social protests were frequent in Chinese history and played an important role in politics. In fact, almost every dynasty's collapse in Chinese history was linked to violent protests in either a direct or indirect way. Poverty and hunger were the primary causes of such protests, that mostly took the form of peasant uprisings (Long, 2016; Wang, 1884). Perry (1980) suggests that violence and banditry were a survival strategy for desperate farmers, especially in regions hit by bad harvests. Uprisings were also linked to the unrestrained behavior of local elites and government officials, who relentlessly exploited the local peasants. The local elites, powered by their economic and social superiority, usually encroached on the private lands of peasants and profited from the manipulation of the local silver-money exchange rates to exploit them. Local officials, who were usually in debt due to the large investment they had to make to prepare for the Keju exam, did not hesitate to levy additional taxes on peasants (Miao et al., 2021).

The peasant revolutionaries' dissatisfaction toward the local elite and officials was visible in their political slogans. For instance, during the Li Zicheng revolution at the end of the Ming dynasty, peasants raised the slogan of "equalize land ownership, exempt tax and debt of the poor" to express their anger toward over-taxation by the government, and toward exploitation by the local elite. Similar slogans were also used in the Taiping revolution during the Qing dynasty period.

Riots and uprisings of discontent peasants were justified by Confucian classics, especially those from Mencius (Ouyang, 2009). From Mencius' point of view, there is no difference between ordinary people and rulers, and ordinary people could become legitimate rulers if they treat people with mercy ("Everyone can be a saint", Mengzi, 5A6, 6B2). This idea justified Chinese people's enthusiasm for uprisings. The leaders of peasant uprisings were often failed Keju examinees, probably due to their higher education and advantage in effective communication. However, the history of China is also characterized by revolutionary leaders coming from the lowest peasant class with almost no education, such as the founder of the Ming dynasty, Zhu Yuanzhang.

[10]It is interesting to note that the Qing emperors did not try to dispel the superstitious interpretation of solar eclipses. For example, the Qing's court stipulated that during predicted eclipses, all officials had to participate in a ritual to "save the sun". Officials in the capital were required to join the ritual at the gate of the Ministry of Rites, wearing white clothing (Shi and Lyu, 2002).

## 2.3 The Fall of the Qing dynasty

The Qing dynasty officially ended in 1912, but the abdication of the last emperor was preceded by a decade of increasing discontent and social protests. In the early 20th century, the Qing government was struggling with the age-old "king's dilemma" of how to modernize a backward society – such as China at the time – while yet remaining in power (Huntington, 2006). In the last decade of the dynasty, the central government's authority had weakened after a series of military defeats, most notably the Sino-Japanese War and the Boxer Rebellion, and was saddled with large war reparations.

A set of reforms were hurriedly put into place by the ruling dynasty, sending China spiraling out of control. Some of the reforms resulted in appointments that gave the royal family total control over key political positions, leaving the population dissatisfied. Talented young Chinese were sent to study abroad, but when they returned, they brought new revolutionary ideologies with them. The military started recruiting educated individuals, leading to a new army full of young literate soldiers whose opinions were influenced by revolutionary ideas and who resented the regime for the elimination of the civil service examinations. A series of changes aimed at establishing new institutions, such as the police, greatly increased the local fiscal burden. Local governments, which were not restrained by a weakening central government, took the opportunity to levy arbitrary taxes, which substantially increased public resentment (Hao et al., 2022). Inflation and high taxes started to force companies out of business, increasing unemployment.

Ultimately, social tensions during the first ten years of the 20th century translated into a wave of social unrest episodes, which were initially repressed by the government. Many such episodes were started by merchants, workers, and farmers in response to deteriorating economic conditions, exploitation by local elites and discontent at the government's policies (Wang, 2008; Yang, 2010). In October of 1911, revolutionaries in Wuchang were forced into an uprising as their plan of action was exposed due to a gunpowder explosion. Their action was successful and was quickly supported by other regions, leading to 15 out of 24 provinces declaring independence within a month. The Qing dynasty's rule was overturned four months later, in February of 1912.

# 3 Data

## 3.1 Solar eclipses

Data on solar eclipses is sourced from Xavier M. Jubier's website Five Millennium (-1999 to +3000) Canon of Solar Eclipses Database, compiled from NASA Technical Publication TP-2006-214141. The database provides detailed GPS information on all solar eclipses that occurred in the last 5,000 years, including the timing and the magnitude of each observable eclipse in each region on earth.[11]

We focus on eclipses whose zones of totality intersect the China territory. Figure 1 reports, as an example of the raw data, the zones of totality for two eclipses, one occurred in 1742 and the other in 1760, that were visible from China. We define county borders during the Qing dynasty period using data from the China Historical Geographic Information System (CHGIS) Version 6 (CHGIS, 2016).[12]. Figure A.1 plots all counties and provinces in CHGIS V6.

We combine the CHGIS data on county borders and NASA information on solar eclipses to generate a panel dataset at county-year level reporting the magnitude of visible solar eclipses in each county and year. A county is considered in the totality zone of an eclipse if the county coordinates reported in CHGIS V6 are within the totality zone of a solar eclipse.

Figure 2 shows the geographical distribution of counties that were in the totality zone of solar eclipses during the Qing dynasty period, along with the number of occurrences of such events (between 0 and 4). Figure 3 shows in blue bars the number of counties in the totality zone of a solar eclipse by year between 1647 and 1911.

## 3.2 Social protests

Data on peasant uprisings during the Qing dynasty period is manually extracted from the Chronology of Warfare in Dynastic China (The Chinese Military History Editorial Team, 2003). This collection includes all wars, conflicts and revolts in the history of China. We focus on the Qing's period (1644-1912). We manually analyze all the records and include in our dataset only peasant uprisings. Figure A.2 shows one example of such records. This

[11] Xavier M. Jubier's website provides GPS Exchange format files that include the northern limit line, southern limit line and central line of each eclipse.

[12] Shapefiles of the Qing dynasty are from China Historical GIS Version 6 (CHGIS V6). The dataset is publicly available and was accessed on Jan 31, 2025. The version used in this project is 1911 Layers UTF8 Encoding. CHGIS V6 has two separate files: county coordinates and county polygons. We focus on China Proper, and remove the following provinces from the Qing China shapefiles: Heilongjiang, Jilin, Fengtian, Neimenggu, Qinghai, Xizang, Xinjiang, Wuliyasutai, Taiwan (Taiwan was annexed by the Empire of Japan in 1895).

example describes a revolt taking place in 1813. The record provides information on the time, location, cause, name of the leader, and the total number of participants in the revolt. It also provides a brief description of how the revolt developed and the interactions between the rebels and the government.

The information used to compile the Chronology of Warfare in Dynastic China comes from both the central government and local chronicles. For most large-scale rebellions, there exists a very detailed central government record. The central government had a strong incentive to be informed of local rebellions in a timely fashion, because if they did not tackle such issues promptly, small insurgencies could turn into nation-wide rebellions. Therefore, local officials were required to report information about local rebellions to the central government. The failure to do so could lead to the removal of a local official.

The local chronicles, which record all major local events occurring in a given county, also recorded such insurgency actions. Most authors of the local chronicles were independent, reputable local intellectuals, who were not under the scrutiny of the government (Almond et al., 2019). Their relative freedom mitigates concerns that news relating to local insurgencies would be intentionally omitted.

Figure 3 shows in gray bars the number of counties reporting at least one peasant uprising against the government in each year during the period between 1647 and 1911 according to the Chronology of Warfare in Dynastic China. As shown, peasant uprisings were relatively frequent during the Qing dynasty period. The peak in the number of unrest episodes in the 1850s and 1860s is due to the Taiping rebellion (1851-1864) and the Nian rebellion (1853-1868). The Taiping rebellion in particular was the largest peasants' uprising in Qing China, with estimated casualties of between 30 and 50 million (Xu and Yang, 2018). Figure 4 (a) reports the geographical distribution of peasant uprisings recorded during the Qing dynasty period according to the Chronology of Warfare in Dynastic China.

We source data on protest events leading to the fall of the Qing dynasty from "The chronology of civil protests in the Late Qing Dynasty" (Ding and Zhang, 1982), which documents protest events that occurred between 1902 and 1911. The original source of information on protest events used in Ding and Zhang (1982) is national newspapers – including the Universal Gazette, Shun Pao, and Ta Kung Pao – and government archives. The data covers 1,511 protests in the final 10 years of the Qing dynasty, providing information on the date, location, and a short description for each event. A potential limitation of this data source is that newspapers and governments' reports tend to overlook riots that occur in border provinces and minority regions, potentially leading to omissions. We think this issue is attenuated in our empirical analysis by the use of province fixed effects, which absorb heterogeneity in coverage across Chinese provinces. See Hao et al. (2022) for an application of

this data to the study of the effect of the abolition of the civil service exam on the incentives to misbehave of local elites. Figure 3 shows in red bars the number of rebellions between 1902 and 1911 sourced from Ding and Zhang (1982). Figure 4 (b) reports the geographical distribution of the rebellions.

Data on social protests in present-day China is sourced from the local unrest database of the Chinese Academy of Social Sciences (CASS database hereafter, 2023). This database is collected by the Institute of Law of CASS, the top research institute and think tank in China. The database is designed to record all social unrest events involving more than 100 participants. It contains 916 observations taking place between 2001 and 2013 in Mainland China (Hong Kong SAR and Macao SAR excluded). All observations come from the coverage of newspapers published in mainland China and major online websites. For each unrest observation, the database reports detailed information on the location, the type of protest, the type and number of participants, the aim and duration of the protest event.

We complement CASS with three additional data sources that have been used in the recent literature on social unrest in China: the CLB, ACLED and GDELT. The recently dissolved China Labour Bulletin (CLB, 2023) was a Hong Kong-based non-governmental organization that maintained a dataset recording labor disputes and collective actions involving workers across mainland China. CLB aggregates reports from social media, news outlets, and direct worker submissions, and focuses specifically on labor-related unrest such as strikes, wage arrears protests, and factory disputes. As CLB reports the number of participants in each protest, we filter events with at least 100 participants to match the CASS dataset. We also rely on the Armed Conflict Location and Event Data Project (ACLED, 2024). This dataset draws on largely overlapping sources as the CLB for protest events in mainland China. Finally, we use the Global Database of Events, Language and Tone (GDELT, 2013). This is an event database designed to monitor news media worldwide and uses automated coding to extract political events, including protests.

Figure A.3 reports the aggregate monthly number of protest events reported by the four datasets used in our analysis. For all datasets we restrict to protests located in the "China proper" provinces. Data sources differ in their time coverage and in their level of geographical resolution. We summarize this information in Table A.1. CASS reports the county of the protest for each event. This is the same level of variation as our main results on the Xinhai revolution. CLB reports the name of the province and prefecture of each event. GDELT reports geographical coordinates that, for the majority of events, correspond to the capital of the province or of the prefecture where the event occurred. As can be seen in Figure A.3, ACLED has a large degree of overlap with CLB. It reports geographical coordinates that

can be mapped to either the county or the prefecture of each event. Because CASS is the only dataset for which geographical location of protest events is consistently reported at the county level, we use CASS for the county-level analysis, and all datasets for the prefecture-level analysis.

## 3.3 Data on martyrs, temples and legends

To explore the role of local anti-Qing memories over the long run, we collected data on temples and legends celebrating officials who died in the war against the Qing invasion (1644-1664). The list of names of anti-Qing martyrs comes from the "Emperor Designated Book of Pre-dynasty's Martyred Officials"(Wuying Hall Scholars, 1776). This book was compiled and distributed by the Qing rulers. This is because the Qing court, as an ethnic-minority ruler, constantly aimed at inspiring trust and loyalty toward the regime among its subjects and to establish its own ruling legitimacy.[13] The book contains a list of 3,787 civil and military officials who died in the war against the Qing's invasion. The officials' names are ordered according to the princes they served and died with. Figure A.4 reports an example of an entry of the book, which mentions an official named Jie Chongxi who martyred with the Right Prince Lu.

As the actions of these early martyrs were described in a book issued by the Qing court, these rebels were officially recognized by the government. As a result, it was legitimate for the subjects to construct temples dedicated to them ("Martyrs' Shrine"), publicly worship them and tell their stories. On the other hand, worshiping and celebrating the leaders of peasant uprisings that happened during the Qing dynasty period may have triggered the intervention of the government. Over time, the temples dedicated to early anti-Qing martyrs became a symbol and a place to worship *all* anti-Qing martyrs and rebels, whether they were recognized by the government or not. This is consistent with the Chinese tradition of expressing discontent about the government in indirect ways.[14]

We extracted information on the location of temples dedicated to such martyrs from local county chronicles. Local chronicles contain a section ("geographic" section) whose main purpose is to report the location of the important sites of a given county. Temples were important not only because of their religious and cultural purposes, but also because the periodic markets held outside temples were usually the busiest ones and the most important gathering opportunities for rural residents in China. Figure A.5 provides an example of the

[13]Interestingly, the Qing court compiled another book, called "Double-faced Ministers", at the same time. It collected all pre-dynasty's ministers who betrayed their own regime and served for the Qing court.

[14]In a famous case that happened in 1661, Chinese intellectuals cried in a Confucian temple as a way to express their anger towards the government's over-taxation.

description of a martyrs' temple as reported in local chronicles. As shown in the example, county chronicles usually provide the name of the temple, the location of the temple and a brief description of the individuals worshiped in the temple.[15] We collected information on 720 temple records in 428 counties. Figure 5, panel (a), shows the geographical distribution of such temples.

The accounts of martyrs' actions against the Qing's invasion were also recorded in the "memorabilia" section of local chronicles. This section recorded the important events that took place within the county, including the fight against the Qing's invasion, especially after the Qing court officially recognized the legitimacy of those actions. Notice that the heroic actions of local martyrs were reported in the chronicles even when the actual battle did not take place within the boundaries of the county. Figure A.6 presents an example of an account of a martyr's actions. The chronicle is from the county of Guichi, Anhui province, and provides a detailed description of the anti-Qing struggle of martyr Cao Dagao, who was from Guichi but whose actions mostly took place in the Jiangxi province. His associate and a non-local martyr, Jie Chongxi, was also mentioned in the local account because of his connection with Cao.

Again, we use both electronic and manual searches to collect all the legends of martyred anti-Qing officials in the chronicles. To isolate the effect of the local diffusion of anti-Qing sentiment, we focus in particular on written accounts that describe the rebels' actions with a positive tone. Our final dataset includes 2,515 martyrs' legends recorded in the chronicles of 461 counties. Figure 5, panel (b) shows the geographical distribution of counties with recorded anti-Qing legends in local chronicles.

Appendix A provides further details on the construction of these variables, including the keyword-based procedure used to identify genuine references to martyrs in local chronicles, the geocoding of temple records, and the data sources for county administrative rankings and clan records used in the mechanism analysis. Table A.2 provides a list of all variables used in the analysis, along with their definitions, original data sources, and level of variation. Table 1 reports summary statistics.

# 4 Identification Strategy

Our identification strategy relies on the assumption that the timing and area of visibility of solar eclipses across Chinese counties are plausibly exogenous. This assumption is based

[15] As a sign of respect, temples commemorating martyrs are typically referred to by their posthumous titles or by their acts of loyalty and sacrifice, rather than by their personal names. We hired graduate students from the Department of History to identify temples associated with such posthumous titles or acts of loyalist resistance.

on the fact that solar eclipses are determined by the relative position of the sun, the earth, and the moon, and not by geographic, demographic, or economic characteristics of the area exposed to it on the surface of the earth. We employ two main specifications: a panel approach and a cross-sectional long-differences approach. We present each in turn, together with the identification assumptions they require and the evidence we use to support them.

## 4.1 Panel approach to study short-run effects

To study the effect of solar eclipses on peasant uprisings during the Qing dynasty we estimate the following panel specification:

$$1(rebellion)_{ipt} = \alpha_i + \alpha_{pt} + \beta 1(eclipse)_{ipt} + \theta X_{ipt} + \Lambda_t G_{ip} + \varepsilon_{ipt} \quad (1)$$

The outcome variable $1(rebellion)_{ipt}$ is a dummy equal to one if a peasant uprising was recorded in county $i$, located in province $p$ during year $t$. The independent variable $1(eclipse)_{ipt}$ is a dummy equal to 1 if county $i$ was in the totality zone of a solar eclipse during year $t$. The totality zone of an eclipse is the area on the surface of the earth in which the moon covers the totality of the sun disk. Notice that this measure is based on its theoretical visibility in perfect weather conditions – i.e. in the absence of clouds, fog or smoke – as reported by NASA.

The key identification assumption in this panel specification is that year-to-year variation in exposure to solar eclipses is plausibly random, conditional on fixed effects, which allows us to estimate the causal effect of this celestial event on the probability of rebellions. County fixed effects ($\alpha_i$) absorb all time-invariant county characteristics. Province fixed effects interacted with year fixed effects ($\alpha_{pt}$) capture any common macro shock at the province level. $X_{ipt}$ are county-level time varying controls, while $G_{ip}$ are county-level time invariant controls which we interact with year fixed effects. We describe these controls in detail below. Given the richness of this fixed effects structure, the remaining threats to identification are limited to the possibility that the *timing* of eclipses is correlated with the timing of other local shocks.

Even if the timing and location of eclipses is plausibly exogenous, a potential concern with the interpretation of our results is that the timing of eclipses may be correlated with the timing of other local shocks, such as natural disasters, which themselves might increase the probability of social unrest. Some scholars have argued that solar eclipses can increase the probability and intensity of El Niño and La Niña effects, weather phenomena that can cause extreme weather events (Lin et al., 1999). To investigate this concern, in Panel A of Table 2, we study whether the incidence of total eclipses is correlated with the incidence of four types of natural disasters: earthquakes, droughts, floods, and insect infestations. These

are among the most prominent natural disasters that could have affected the Chinese rural economy at the time, which mostly relied on agriculture. As such, local chronicles kept a detailed record of such events, from which we extracted our data. In addition, local chronicles report if a famine occurred in a given year. Overall, we find no systematic differences in the incidence of reported natural disasters and famines. In our empirical analysis, we show that the magnitude of the estimates is stable when including time varying county-level controls ($X_{ipt}$).

A second potential concern with the interpretation of the results is that eclipses could have been anticipated by the rebels or the central government. Anticipating the shock, local rebel leaders could have timed their coordination efforts in certain years. Similarly, the central government could have triggered preventive measures to avoid potential unrest. As discussed in section 2, during the Qing period the study of astronomy was monopolized by the Qing court via the Bureau of Astronomy. Thus, it is plausible to assume that local leaders of peasant uprisings had no access to the scientific instruments and knowledge necessary to predict the timing of a solar eclipse and focus their coordination efforts accordingly. On the other hand, astronomers of the Qing court could predict the occurrence of solar eclipses. In fact, one of the most important tasks of the Bureau of Astronomy was to produce predictions of solar and lunar eclipses, exactly because of their special role in Chinese political astrology and their deep-rooted interpretation as a negative omen for the regime (Lingfeng, 2007). However, their predictions were often inaccurate by modern standards (Lyu and Shi, 2003b,a). Based on our reading of the literature, we think it is plausible to assume that the Bureau of Astronomy was not able to predict which specific areas within a given province of China would have been under the totality zone during any given solar eclipse. Any aggregate response to a predicted eclipse put in place by the Qing court at the national or provincial level would be absorbed by province fixed effects interacted with year fixed effects.

A standard econometric challenge when exploiting variation across geographical units is the degree of spatial correlation, which might lead to inflated t-statistics. To account for spatial correlation in residuals across Chinese counties, we cluster standard errors at the prefecture level in all our specifications. Prefectures are intermediate administrative units between counties and provinces. In our sample, we have 249 prefectures, with each prefecture encompassing on average 6.2 geographically contiguous counties. Our empirical analysis focuses on the eighteen provinces of what is usually referred to as "core China" or "China proper" under the Qing dynasty. This implies that it excludes the inner domains of Qing China, namely Manchuria, Inner and Outer Mongolia, Xinjiang and Tibet. In the Appendix, we also report Conley-adjusted standard errors (Conley, 2010) at distance cutoffs of 50, 100, 200, and 300 kilometers for all main results, to address concerns about spatial

correlation in residuals beyond the prefecture level.

## 4.2 Cross-sectional approach to study long-run effects

To study the long-run effects of past rebellions on the incidence of the protests that led to the 1911 Revolution and the fall of the Qing empire we use an instrumental variable approach that relies on cross-sectional variation in the incidence of past rebellions generated by eclipses during the Qing dynasty period.

To this end, we start by estimating the predicted incidence of past protests per capita during the Qing period with the following first stage specification:

$$rebellion_{ip,1647-1899} = \alpha_p + \gamma eclipse_{ip,1647-1899} + \theta X_{ip} + u_{ip}, \tag{2}$$

where $rebellion_{ip,1647-1899}$ is the total number of peasant uprisings per capita observed in county $i$ between 1647 and 1899, and $eclipse_{ip,1647-1899}$ is the total number of solar eclipses observed in county $i$ during the same period.[16] Next, we estimate the following second stage specification:

$$rebellion_{ip,1902-1911} = \alpha_p + \beta \widehat{rebellion}_{ip,1647-1899} + \theta X_{ip} + \eta_{ip}, \tag{3}$$

where $rebellion_{ip,1902-1911}$ is the total number of protests per capita observed in county $i$ between 1902 and 1911. Variables in per capita terms in the first and second stage regressions are computed using estimates of county-level population from Skinner et al. (2011), which refer to population in the capital city of each county estimated as of 1893.[17] All specifications include province fixed effects ($\alpha_p$) and a set of county controls described below. Standard errors are clustered at the prefecture level.

The identification assumptions for the cross-sectional approach are more demanding than for the panel. Differently from the panel specification, this approach does not include county fixed effects and therefore cannot absorb time-invariant unobservable county characteristics. The key assumption is that the cumulative number of solar eclipses experienced by a county over roughly 250 years of Qing rule is uncorrelated with other determinants of protest participation in the 1911 Revolution, conditional on province fixed effects and observable county

[16] The Qing dynasty period lasted from 1644 to 1912. We exclude the years 1644 to 1646 in which the Qing consolidated their power. We exclude the years 1900 to 1912 as they are characterized by the wave of protests that led to the fall of the Qing which we use as our main outcome variable.

[17] This dataset classifies each county's capital city into one of 11 population size categories, ranging from up to 500 individuals to above 500,000. We assign to each county the midpoint of the population size category of its capital city.

controls. Although we cannot fully rule out violations of this assumption, we present evidence that supports it.

*Spatial variation in eclipse probability.* The astronomical literature has shown that eclipse probability is not perfectly uniform across locations on Earth. In particular, Meeus (1982) documents that eclipse probability varies systematically with latitude, with eclipses being relatively more frequent in the northern hemisphere.[18] We find evidence that eclipse probability varies with latitude also in our sample period. This could be problematic if other determinants of persistence in protest are also correlated with latitude. To address this, we control for latitude and for the frequency of eclipses in each county between 2000 BCE and 3000 CE (excluding the Qing period) in all cross-sectional specifications. The inclusion of the long-run eclipse frequency as a control ensures that our estimates are not driven by the spatial distribution of eclipse probability across Chinese counties, but rather by the specific timing of eclipses during the Qing period.[19]

*Balance on observable characteristics.* In Panel B of Table 2, we regress the standardized number of eclipses during the Qing period on observable county characteristics, controlling for province fixed effects. Column (1) reports coefficients from separate bivariate regressions for each characteristic. As shown, eclipses are more frequent at higher latitudes also in our sample, consistent with the astronomical literature. We also find significant correlations between Qing-era eclipse exposure and the log of agricultural land tax per capita, suitability to sweet potato cultivation, longitude degrees, distance to the coastline, and the number of eclipses outside the Qing period[20]. By contrast, we find no significant differences in all other observable characteristics, including population, number of local clans (Dincecco and Wang, 2021; Bai et al., 2023), soil suitability for rice and millet, terrain ruggedness (Nunn and Puga, 2012), distance to Beijing, distance to the provincial capitals, or distance to the Yangtze river. In column (2), we include all characteristics jointly. The joint F-statistic is 4.05, indicating that observable characteristics do have some explanatory power for Qing-era eclipse exposure.

The existence of these correlations is a common concern in studies that investigate the

[18] Additionally, Buus (2014) shows that the duration of total solar eclipses tends to be longer closer to the equator.

[19] Notice that extending the eclipse probability calculation beyond this 5,000-year window is not feasible due to increasing astronomical uncertainty: uncertainty in $\Delta T$ (the difference between Earth's actual rotation time and the standardized atomic clock time (Espenak and Meeus, 2006a)) makes it difficult to determine the exact location of totality zones, and long-term gravitational interactions between celestial bodies make precise orbital predictions unreliable beyond this horizon (Laskar, 1989; Lithwick and Wu, 2014; Espenak and Meeus, 2006b).

[20] Data on suitability to difference crops is sourced from Food and Agriculture Organization of the United Nations and International Institute for Applied Systems Analysis (2021), distance to the coast from OpenStreetMap Contributors (2024), distance to main rivers from Natural Earth (2022)

persistent effects of past phenomena using long-run cross-sectional variation (Voth, 2021). We address it by including all characteristics from Panel B of Table 2 as controls ($X_{ip}$) in equations (2) and (3). These characteristics are also included in the panel regressions described in section 4.1, interacted with year fixed effects. We show that the magnitude of coefficient estimates is stable when adding these controls, which lends support to the identification assumption.

# 5 Empirical Results

## 5.1 The Effect of Solar Eclipses on Rebellions in Qing China

We start by studying the effect of solar eclipses on local rebellions during the Qing dynasty period. We estimate equation (1), where the outcome variable is a dummy equal to one if a peasant uprising was recorded in a given county and year. The results are reported in Table 3. The estimated coefficient in column (1) indicates that counties that experience a total solar eclipse have a 12.6 percentage points higher probability of also experiencing a peasant uprising in the same year. In column (2) we include province times year fixed effects, while in column (3) we include the set of county controls described in section 4. As shown, the magnitude of the point estimate on the total eclipse dummy remains stable with the inclusion of these controls.

The premise behind the link between solar eclipses and rebellions is a shared interpretation that solar eclipses are a divine signal of undermined legitimacy. The eclipse-illegitimacy interpretation is mainly held by the Confucian school. Thus, we test whether the effects of eclipses on the propensity to rebel change depending on the local diffusion of Confucianism. To proxy for it, we use the geographical distribution of Confucian temples, under the assumption that regions with Confucian temples had a larger diffusion of Confucian ideology among the local population (Kung and Ma, 2014). In columns (4) and (5) of Table 3 we estimate equation (1) separately in counties with vs without Confucian temples. The effects are present in both sub-samples, with larger coefficient estimates in the sample of counties with Confucian temples. In the last column we formally test for the magnitude and statistical significance of the difference in the effect of eclipses across the two samples. As shown by the interaction term, this difference is statistically significant and its magnitude indicates that the effect of eclipses on the probability of rebellion is about 66% larger in counties with Confucian temples.

We explore the timing of the relationship between eclipses and rebellions with the following dynamic specification:

$$1(rebellion)_{ipt} = \alpha_i + \alpha_{pt} + \sum_{k=-3}^{k=+3} \gamma_k 1(year = k)_{ipt} + \Lambda_t G_{ip} + \theta X_{ipt} + \varepsilon_{ipt} \quad (4)$$

where $1(year = k)$ is a dummy equal to 1 if year $t = k$ for county $i$, and captures the time relative to a total eclipse year in county $i$, which we set at $k = 0$. We include the 3 years prior and the 3 years after the total eclipse. The specification includes county and province times year fixed effects, as in equation (1). The results are reported in Figure 6. The effect of solar eclipses on unrest materializes in the same year in which the eclipse occurs, while the estimated coefficients on leads and lags are small and not statistically different from zero. It is important to emphasize here that unrest events are assigned to the year in which they started according to local chronicles. This might help explain why the documented effect of eclipses on rebellion is mostly contemporaneous.

We also present a set of robustness and additional results in Table A.3. To account for overdispersion in the eclipse variable due to the high number of zeros, we re-estimate our main specification using a Poisson regression model. The results are reported in column (2) and show a positive and significant effect of the total eclipse variable on the rebellion outcome. The magnitude of the estimate implies that total eclipses increase the probability of peasant uprisings by about 16.7 percentage points, which is comparable in magnitude to the baseline result.[21]

In column (3) we re-estimate equation (1) excluding from our sample the years of the Taiping rebellion. Figure 3 shows a high concentration of unrest events around the years of the Taiping rebellion. The rebellion originated in Jintian, a town in Guangxi Province, where Hong Xiuquan and his Han-ethnicity followers declared independence from the Qing central government of Manchu ethnicity. After Hong's army captured Nanjing in 1853 and established it as their capital, the rebellion escalated into a civil war that lasted until 1864, caused millions of casualties on both sides, and ultimately ended with the victory of the Qing government. As shown, the point estimate on total eclipses is unchanged, indicating that the results are not driven by this particular historical episode.

Next in columns (4) and (5) of Table A.3 we split our sample into two sub-periods. We find positive and significant effects in both periods. In terms of magnitude, the estimates indicate that solar eclipses have higher predictive power on peasant uprisings in the early period of the Qing rule than in the period between 1800 and 1911.

Column (6) includes an interaction between the current eclipse indicator and a dummy for whether the county experienced a total solar eclipse in a prior year. The interaction term

[21] This number is obtained by multiplying the incidence rate ratio implied by the Poisson estimate by the average probability of observing a rebellion in non-total eclipse years ($37.2 \times .0045 = 0.167$).

is positive and statistically significant, indicating that the impact of eclipses on rebellions is stronger in counties that already experienced a solar eclipse in the past. This result is consistent with a reinforcement mechanism: in counties where eclipses had already triggered unrest, the shared memory of past collective action amplified the response to subsequent eclipse events. If one interprets the lagged eclipse as an instrument for past protest experience, and the current eclipse as an instrument for the current rebellion opportunity, the significant interaction suggests that communities with a history of eclipse-driven protest are more responsive to new coordination signals.

Finally, in column (7) we interact solar eclipses with the occurrence of other natural disasters (including floods, famines and droughts) to study whether their impact is stronger in periods during which a county is already experiencing a negative economic shock. As shown, the coefficient on the interaction term is positive but not precisely estimated, which is likely driven by the limited number of counties experiencing both shocks in the same year.

As a more direct test that our results are driven by the specific timing of eclipses during the Qing period rather than by spurious spatial correlations, we follow the approach of Madestam et al. (2013) and re-estimate the first stage using placebo eclipse frequencies from periods outside our sample. Using NASA's 5,000-year eclipse catalog, we construct placebo treatments from eclipse patterns of equivalent length shifted outside the Qing dynasty period. If our results were driven by the spatial distribution of eclipse paths, we would expect similar effects for eclipses from other periods.

The results are presented in Figure 7. Panel 7(a) reports the cumulative distribution of estimated coefficients of the effect of eclipses on protests across all placebo windows. As shown, the coefficient from the actual Qing period is larger than all placebo coefficients. Figure 7(b) plots each coefficient against its p-value. As shown, the Qing-period estimate is the one with the largest magnitude and the lowest p-value. These results suggest that the predictive power of eclipses on rebellions is not an artifact of the spatial distribution of eclipse paths, but rather reflects the specific historical conditions of the Qing period.

## 5.2 Legacy of Rebellions and the 1911 Revolution

In this section, we test the long-run effects of a history of peasant uprisings on the incidence of the protests that led to the 1911 Revolution and the fall of the Qing empire. For this analysis, we exploit plausibly exogenous variation in the incidence of past rebellions generated by eclipses during the centuries of Qing rule. This approach and its key identification assumptions are discussed in section 4.2.

We start by estimating the first stage equation (2). The results are reported in Table

4. The estimated coefficient in column (3) indicates that counties with 1 standard deviation larger number of solar eclipses during the Qing dynasty period (0.895) experienced about 0.23 more rebellions per 1,000 inhabitants. The magnitude of the first stage coefficient remains stable when controlling for province fixed effects and when including the time invariant county controls described in section 4.2. The first-stage F-stat is 21.33.

Next, in Table 5, we study the effect of past protests on the incidence of protests during the 1902-1911 period, which hereafter we refer to as 1911 Revolution protests. We measure protests in number of events per capita, and present results for three specifications: the correlation between past protests and 1911 Revolution protests, a reduced form estimate of the effect of total eclipses on 1911 Revolution protests, and the 2SLS estimate obtained estimating equation (3). All specifications include the same set of county controls and fixed effects used in the first stage, and report standard errors clustered at the prefecture level – the intermediate administrative unit between counties and provinces – to account for spatial correlation in the error term.

Column (1) shows that Chinese counties with higher protest per capita during the Qing period also experienced more protest per capita during the 1911 Revolution period. The magnitude of this correlation declines but remains precisely estimated when controlling for county characteristics in column (2). We visualize the correlation in Figure 8. The plotted points are residuals from regressing protests per capita in the two time periods on province fixed effects and county-level controls. The slope of the least squares regression line plotted through the points is 0.82 with a standard error of 0.06. As shown, this relationship is not driven by extreme observations, with most county-level points being close to the regression line.

In columns (5) and (6) we report the 2SLS coefficients. The magnitude of the coefficient implies that counties with a standard deviation larger number of rebellions per 1,000 people during the 1647-1899 period (1.2) had about 1.3 more social protests per 1,000 people recorded between 1902 and 1911. The magnitude of the OLS estimate in column (2) is about 20% smaller than the 2SLS estimate in column (6). This downward bias in the OLS estimation could be driven by several potential sources of endogeneity in the OLS regression. For example, regions in which the Qing court expected protests to erupt might be treated differently by the central government to avoid such rebellions to resurface. Indeed, local officials during the Qing period (magistrates) were judged and promoted by the central government based also on their ability to deal with local crime and violence (Chu, 1962). If the measures taken by the central government and local officials in areas considered more at risk were effective, this would make such regions less likely to display rebellions in the future.

As an additional check on spatial correlation, Table A.4 in the Appendix reports our main

results with standard errors adjusted using the method of Conley (2010) at distance cutoffs of 50, 100, 200, and 300 km. All estimates remain statistically significant at conventional levels across all cutoffs.

### 5.2.1 Half-life of persistence

A natural question is whether the effects of past rebellions on participation in the 1911 Revolution depend on how far back in time such rebellions occurred. To investigate this, we divide the Qing dynasty period into three sub-periods of roughly equal length (1647-1730, 1731-1814 and 1815-1899) and estimate separate reduced form specifications for each. In addition, we test whether eclipses that occurred before the Qing period predict 1911 Revolution protests, using both a late-Ming window (1563-1646) and the entire Ming dynasty (1368-1646).

The results are reported in Table A.5. As shown, eclipses from the most recent sub-period (1815-1899) and the middle sub-period (1731-1814) have positive and statistically significant effects on 1911 Revolution protests, with point estimates of 0.377 and 0.440 respectively. In contrast, eclipses from the earliest Qing sub-period (1647-1730) have a smaller and statistically insignificant coefficient. Eclipses from before the Qing dynasty have small, negative, and statistically insignificant coefficients. These results indicate that the memory of past rebellions depreciates over time: the effects of rebellions occurring more than about 180 years before the 1911 Revolution had largely dissipated by the time the protests erupted.

Table A.6 in the Appendix confirms this finding by including Ming-era eclipse exposure directly as a control in the reduced form specification. As shown, the coefficient on Qing-period eclipses remains positive and statistically significant, while eclipses from the late Ming (1563-1646) and the full Ming dynasty (1368-1646) have no significant effect on 1911 Revolution protests.

# 6 Mechanisms

Why should the incidence of past rebellions have long-run effects on the inclination of a community to protest? In this section, we discuss and test several potential channels. First, we discuss the role of *collective memory* of anti-Qing resistance preserved via temples and legends commemorating past rebel leaders in local chronicles. Second, persistence could operate through the *governance response* of the Qing state to local protests. In particular, rebellion-prone counties considered more difficult to govern may have received different administrative treatment in the long run, reinforcing conditions for future unrest. Finally, past

rebellions may have strengthened local *social capital*, for instance by fostering clan networks that facilitated future collective action.

## 6.1 Collective memory via celebration of early anti-Qing leaders

The commemoration of selected persons has been documented in many cultures as a way of shaping the collective memory and forging the cultural identity of a given community.[22] One anecdote that speaks to the potential role of commemorating individuals of the past in shaping the future actions of a community is the childhood story of Sun Yat-sen, one of the leaders of the 1911 Revolution. Sun Yat-sen was born in a village close to the hometown of Hong Xiuquan, the leader of the Taiping rebellion against the Qing rule that took place in the 1850s. When Sun was young, his favorite passtime was to listen to the story of Hong Xiuquan told by a Taiping veteran living in his village. In fact, he admired Hong so much that he named himself "Hong Xiuquan Jr".

We use two measures of local collective memory, constructed from county chronicles and described in detail in Section 3.3. The first is the presence of physical temples or memorials dedicated to anti-Qing martyrs. As discussed in Section 3.3, these structures originated from officially sanctioned commemorations but over time became focal points for broader anti-government sentiment. The second is the presence of written accounts in local chronicles that describe the actions of anti-Qing martyrs in a positive tone, which we refer to as "legends." We think of both measures as capturing the ability of a community to codify and transmit a culture of anti-government collective action across generations.

We test empirically whether eclipse-driven past rebellions predict the presence of these markers of collective memory. The results are reported in Panel A of Table 6. We find that counties with a standard deviation larger number of past rebellions per capita – instrumented by eclipses – are 18 percentage points more likely to have legends commemorating anti-Qing martyrs in local chronicles (columns 1-2), 25 percentage points more likely to have physical temples or memorials celebrating anti-Qing leaders (columns 5-6), and have on average 0.42 more anti-Qing martyrs mentioned in local chronicles (columns 3-4). All estimates are statistically significant at conventional levels. These results provide evidence that eclipse-driven past rebellions left a lasting imprint on local culture through the creation of physical and narrative markers of collective memory.

[22] See, for example, Marschall (2009) on the role of memorials in South Africa post-apartheid.

## 6.2 Governance and Administrative Response

A second potential channel through which past rebellions could generate persistence is the administrative response of the Qing state. To explore this channel, we focus on the county classification system implemented by the Qing government in the early 18th century. In an effort to rationalize the personnel appointment system, Emperor Yongzheng (1722–35) commissioned provincial officials to evaluate the governing challenges of each county. Around 1728, Jin Hong, a provincial commissioner, proposed classifying counties using combinations of four post designations to better match officials to local challenges (Liu, 1993; Koss, 2017; Wang and Zhang, 2023). Counties were classified along four dimensions: *chong* (strategically located for transportation), *fan* (administratively complex in public affairs), *pi* (problematic tax collection), and *nan* (difficult due to violent customs and prone to crime). The system was formally adopted in 1732, and the county classifications crystallized by the mid 1700s, with very few adjustments thereafter. In fact, the literature has emphasized the rigidity of the system after this initial period, noting that it created a long-run mismatch between official designations and actual governing challenges. As Koss (2017) notices: "Once formalized, priorities became sticky and hard to change. Frozen in time, the institution took on a life of itself" (p.168).

We argue that, if counties more prone to rebellions were initially classified as more difficult to govern, this administrative labeling could have reinforced the conditions for future unrest. Indeed, the core practical consequence of the ranking was that counties that rated highly on multiple designations were filled through discretionary appointment by provincial governors, rather than the standard lottery system administered by the Ministry of Personnel. Strategically important and potentially difficult counties would then receive officials and resources (both administrative and military) specifically selected to deal with these issues.

We test this hypothesis by estimating the effect of protests in the period 1718–1747 – the decades immediately before and during the formation of the classification system – on the actual classification of each county. This window captures rebellions that provincial officials could plausibly have observed when assessing local governing challenges, while predating the point at which designations became largely fixed. We instrument these early rebellions with eclipses in the same period. The results are reported in Panel B of Table 6. The 2SLS estimates show that eclipse-driven past rebellions significantly predict a county's overall importance level. Decomposing by category, the strongest effects are on the *pi* dimension – difficulty collecting taxes (column 4) – and the *nan* dimension – prone to violence and crime (column 5). These results suggest that rebellions in the first half of the 1700s shaped how the Qing state classified and governed counties in the long run. In turn, this administrative

response may constitute an additional channel of persistence.

## 6.3 Social Capital and Clan Density

A third potential channel through which past rebellions could generate persistence is social capital. Clans (*zongzu*) have long been a central form of social organization in rural China, providing mutual insurance, coordinating public goods provision, and serving as a vehicle for collective action (Dincecco and Wang, 2021; Bai et al., 2023). Past rebellions may have strengthened local clan networks by fostering solidarity among families who shared the experience of collective resistance, creating organizational infrastructure for future mobilization.

We test this hypothesis by examining whether eclipse-driven past rebellions predict the local density of clans, using data from the Chinese Genealogy Knowledge Service Platform organized by Shanghai Library. The results are reported in Panel C of Table 6. We consider three outcome variables: a dummy for whether any clan is recorded in a county (columns 1-2), the number of clans (columns 3-4), and clans per 1,000 people (columns 5-6). Across all specifications, the 2SLS estimates are either negative or small and statistically insignificant, providing no evidence that eclipse-driven past rebellions strengthened local clan networks. An important caveat in interpreting these null results is that clans in China are deeply rooted institutions whose origins typically predate the Qing dynasty by centuries. Unlike temples and legends, clan structures were largely inherited from earlier periods.

## 6.4 Effect of past rebellions on economic development

An alternative channel behind the observed persistence is that higher intensity of social protests in the past might affect local economic development in the long run. If past rebellions depressed local growth – for example, by deterring private or government investment due to higher social instability – the resulting lower economic development could independently increase the propensity to protest by reducing the opportunity cost of participation (Campante and Chor, 2012). This channel would imply persistence through material conditions rather than through cultural or institutional transmission.

We test this channel by estimating equation (3) using proxies of economic development in the early 1900s, before the beginning of the wave of protests leading to the Xinhai revolution. Measures of economic development in the early 1900s are scarce. We rely on three measures that are observable in our data at the county level: the log of county population observed in 1900 from Hu (1935) – available for 719 counties –, the cumulative investment by the central government in new infrastructure located in a given county during the Qing period up to

1900, and the cumulative number of years in which the taxes collected in a given county were lower than the fixed quota established by the Qing court.[23]

The results, reported in Table A.7, show that the 2SLS coefficients are uniformly small and statistically insignificant across all three outcomes. This suggests that an economic development channel is unlikely to be a primary driver of the persistence documented in this paper.

The mechanisms we study can be broadly mapped onto two types of persistence. The collective memory and the social capital channels operate through *people-based* persistence. For example, local narratives of anti-Qing resistance are transmitted across generations within a community, shaping attitudes toward government authority over time. On the other hand, the governance channel and economic development channel operate through *place-based* persistence. For example, the initial Qing administrative classification of a county as prone to unrest or problematic tax collection is a label attached to a location, which shaped the type of officials appointed there and the size of local government regardless of the specific individuals living in the county. The results in Table 6 provide evidence consistent with both types of persistence, at least in the form of markers of collective memory and different central administration that may have affected local grievances and the likelihood of future unrest.

# 7 Legacy of Rebellions and Protests in Present Day China

We conclude the paper by investigating whether the long-run effects of a historical tradition of social unrest extend to present-day China. Between the end of the Qing dynasty and the first decade of the 21st century, China experienced major economic transformations, including rapid economic growth and structural change accompanied by significant internal migration from rural to urban areas and from the interior to the coast. Internal migration is likely to attenuate the role of a historical tradition of anti-government protests on current

[23]Data on infrastructure and tax collection is sourced from "Routine Memorials" – reports made by the six major departments of the central government in the Qing dynasty to the emperor. The six departments included the Ministry of Personnel, the Ministry of Revenue, the Ministry of Rites, the Ministry of War, the Ministry of Justice, and the Ministry of Works. We focus on the Ministry of Works and the Ministry of Revenue. The Ministry of Works was in charge of infrastructure construction for the central government. In their reports, they detailed information on infrastructure construction financed and conducted by the central government, including date, location, and type of infrastructure constructed, such as a temple, a bridge or a road. The Ministry of Revenue was responsible for the central government's financial budget, including the collection of tax quota established by the Qing court for each county. In their reports, they provided detailed information on whether each county in China met its tax quota in a given year.

behavior. On the other hand, the long persistence of such traditions is consistent with certain traits of Chinese society, such as the strong social and cultural linkages that migrants maintain with their place of origin, and the justified right to protest against local authorities perceived as exploitative, which is deeply rooted in the Confucian tradition.[24]

We measure the propensity to protest in present-day China using four different data sources, which vary in their coverage and geographic resolution. Our main source is the Chinese Academy of Social Sciences (CASS) database, which records social protests with at least 100 participants at the county level between 2001 and 2013. We complement this with the China Labour Bulletin (CLB), the ACLED (Armed Conflict Location and Event Data Project) and GDELT (Global Database of Events, Language and Tone). We describe these data sources in detail in section 3.2. We focus on two outcome variables for each source. First, the average number of protests per million people (population observed in 2000) in a given location. This is an average across years in which the data is available. Second, a dummy capturing the occurrence of any protest in a given location during the period covered by a given dataset.

The results are reported in Table 7. Panel A presents county-level results using the CASS database. We find evidence of positive effects that are small in magnitude and marginally significant. Counties with a standard deviation larger number of past rebellions per 1,000 people experienced around 0.05 more protests per million people per year between 2001 and 2013 (column 1). The point estimate in column (2) indicates these counties are 15 percentage points more likely to have experienced any social unrest between 2001 and 2013. In columns (3) to (6), we separate protests by target. The effects are mostly driven by protests targeting local government, with a marginally significant effect on the binary outcome.

Panels B and C report prefecture-level results for all data sources.[25] As discussed in section 3.2, CASS is the only dataset that reports protest events consistently at the county level. Thus, when comparing all four data sources, we aggregate data at the prefecture level. As shown, we find positive effects on both protests per capita and the probability of experiencing any protest across all four data sources, although with varying magnitudes and levels of significance.

Overall, although coefficients are noisier than those estimated for the 1911 Revolution, the results in Table 7 show a broadly consistent picture across all different data sources and

[24]One reason to expect persistence despite migration is China's *hukou* system, which ties access to social services, schooling, and property rights to one's registered place of household origin. Even migrants who live and work in cities typically retain their *hukou* registration in their home county for years or permanently, maintaining strong economic and social incentives to preserve community ties to their county of origin.

[25]We use United Nations Office for the Coordination of Humanitarian Affairs (2020) to map counties to prefectures.

levels of aggregation: areas with more eclipse-driven past rebellions tend to display higher protest activity in the first two decades of the 21st century.

# 8 Concluding Remarks

This paper documents that plausibly exogenous differences in the incidence of past protests can have a persistent effect on the tendency to protest of a given community in the long-run. In particular, we document that Chinese counties with a higher incidence of peasant uprisings against local government officials during the Qing dynasty period (1644-1912) had a higher participation in the protests that led to the 1911 Revolution and the fall of the imperial rule. We provide evidence indicating that the cultural transmission of the past protest experience via the celebration of anti-government leaders is a potential mechanism behind the documented persistence. This suggests that even the memory of events not directly experienced by an individual, but that are transmitted across generations within a given community, can act as a reference point for future behavior by members of that community.

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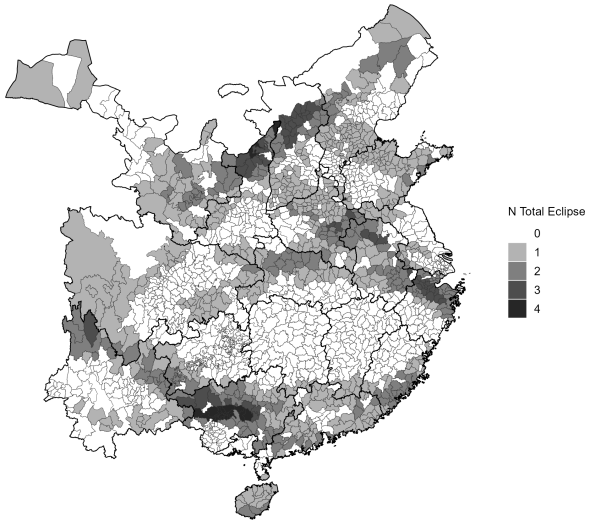
# Figures and Tables

## Figure 1: Zone of totality of solar eclipses: Examples



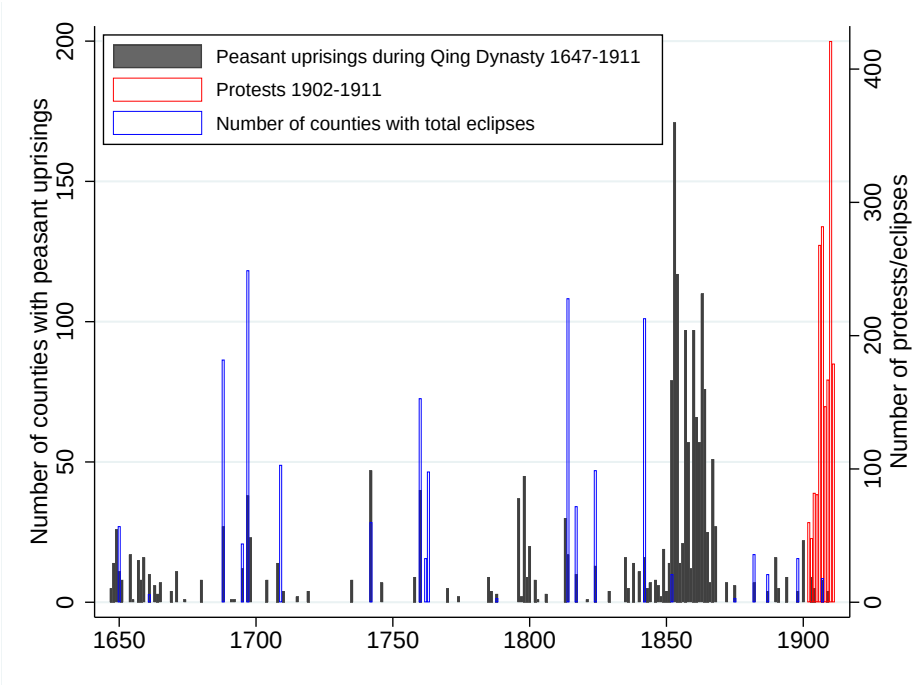
**Notes:** This figure shows the zones of totality of two eclipses: 1742 (left) and 1760 (right). Source: authors' calculation using data from NASA (Espenak and Meeus (2006b)).

## Figure 2: Number of Total Eclipses by County: 1647-1911



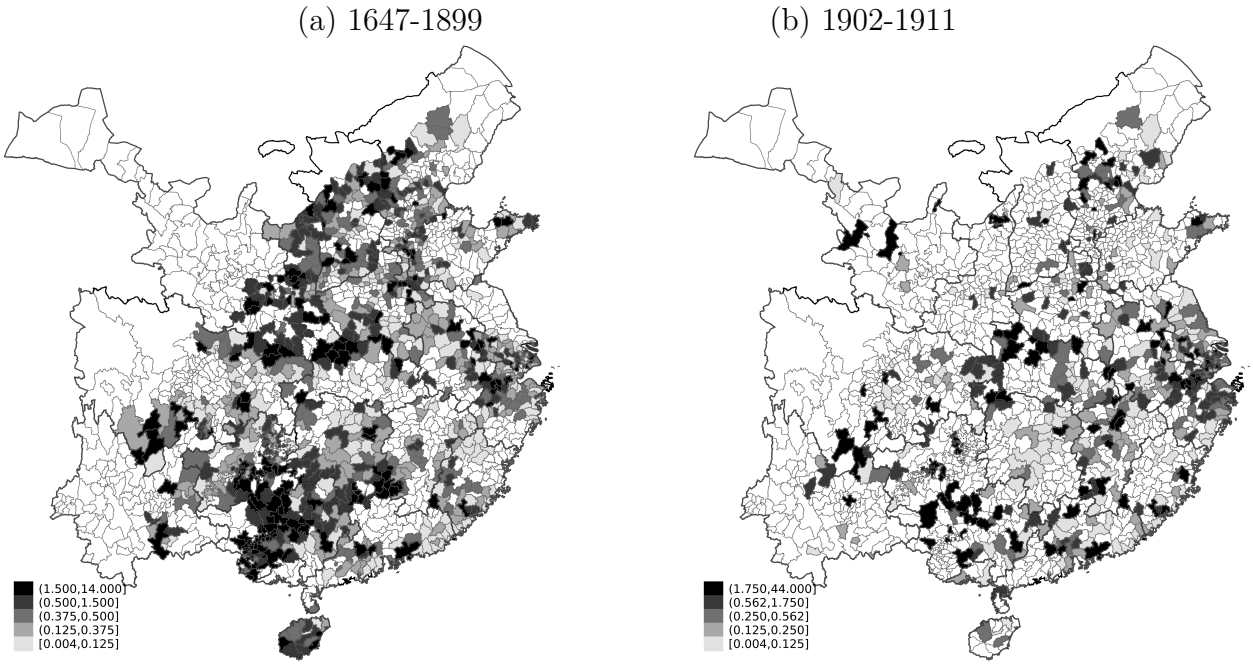

**Notes:** This figure shows the number of total eclipses that each county has observed during the Qing dynasty period. Source: authors' calculation using data from NASA (Espenak and Meeus (2006b)).

## Figure 3: Rebellions and Total Eclipses in China (1647-1911)




**Notes:** This figure show the evolution over time of protests and eclipses. The blue hollow bars show the number of counties experiencing total eclipses in each year during the Qing dynasty, which is built by the authors' calculation using data from NASA (Espenak and Meeus (2006b)). The dark gray bars show the total number of counties reporting peasant uprisings between 1647 and 1911 sourced from The Chinese Military History Editorial Team (2003). The red hollow bars show the number of protests between 1902 and 1911 sourced from Ding and Zhang (1982).

## Figure 4: Geographical distribution of rebellions per 1,000 people

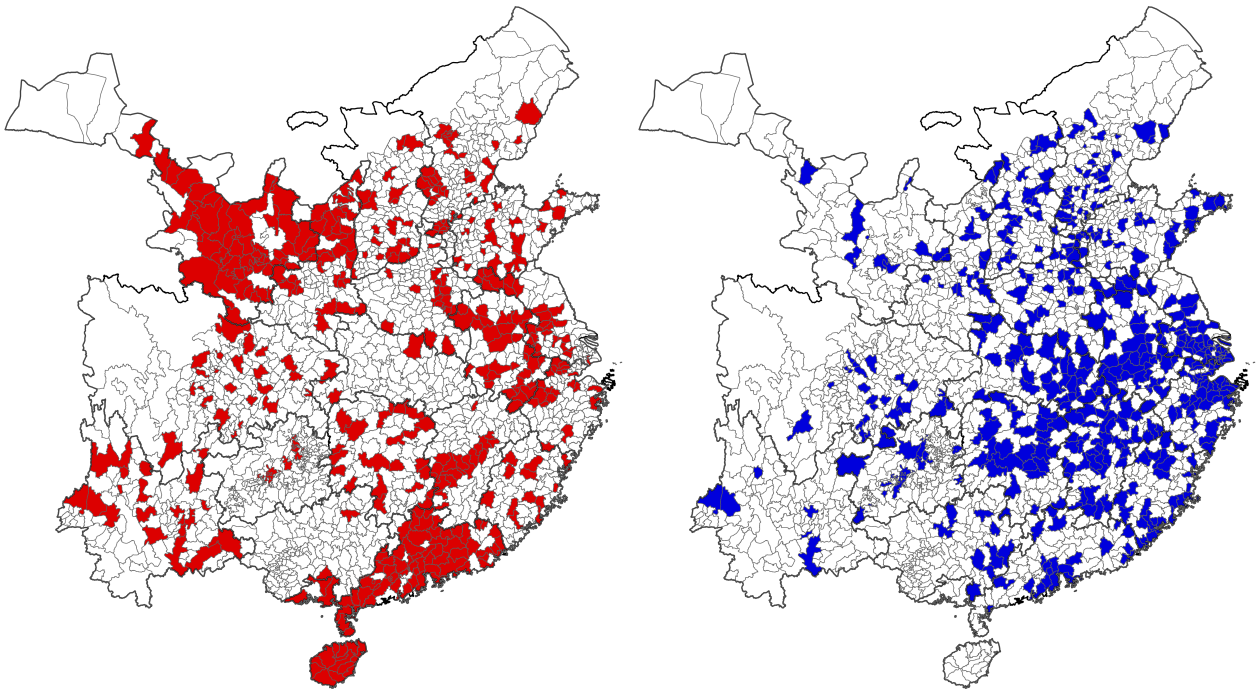



**Notes:** Panel (a) shows the total number of rebellions per 1,000 people that each county has experienced between 1647 and 1899. Panel (b) shows the total number of Xinhai-related rebellions per 1,000 people that each county has experienced between 1902 and 1911. The maps reports county borders (light gray) and province borders (dark gray).

## Figure 5: Geographical distribution of Anti-Qing Memory

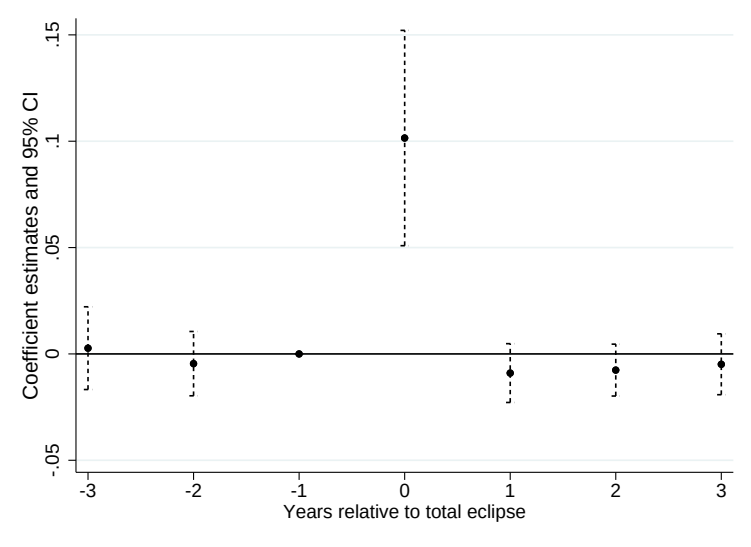
(a) temples

(b) legends



**Notes:** Panel (a) shows the geographical distribution of counties with at least one temple celebrating anti-Qing martyrs. Panel (b) shows the geographical distribution of countries with at least one legend celebrating anti-Qing martyrs in local chronicles.

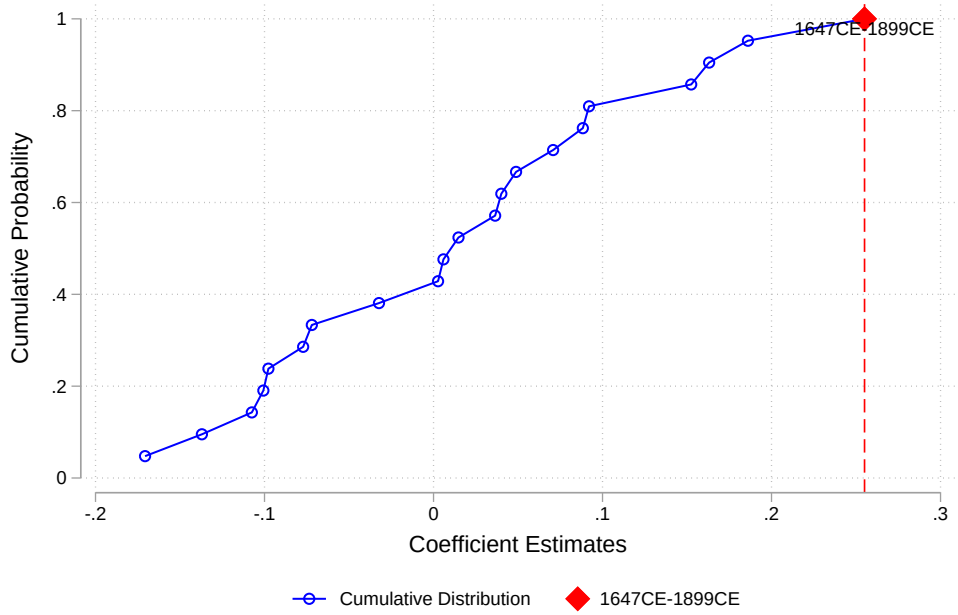
## Figure 6: Timing of the effect of eclipses on probability of rebellion



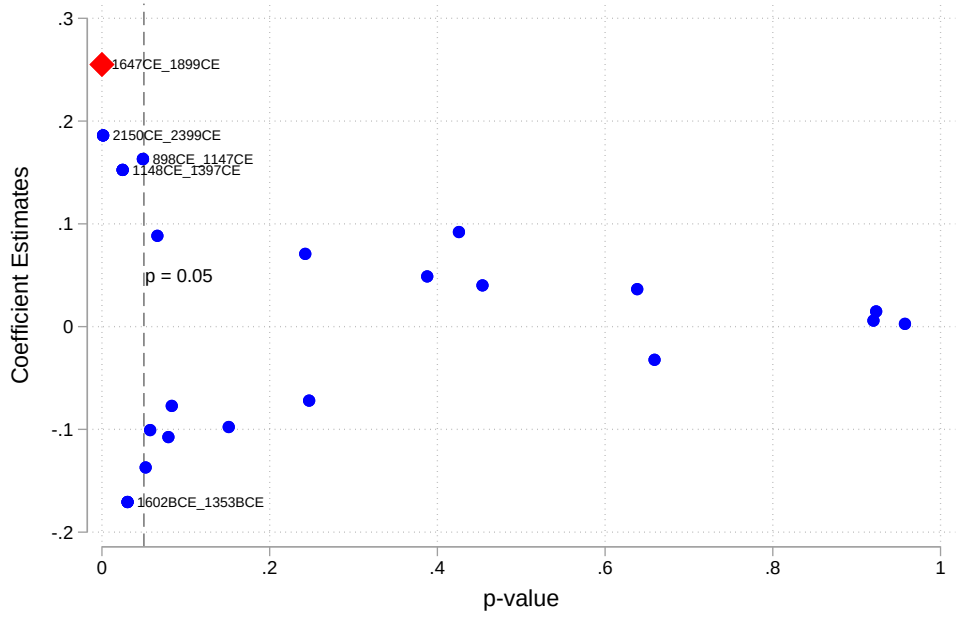

**Notes:** The figure reports the estimated coefficients and 95% confidence intervals on the up-to-3-period leads and lags of the effect of total eclipses on rebellions described in equation (4). Standard errors used to construct confidence intervals are clustered at the prefecture level.

## Figure 7: Placebo Tests - Eclipses from 2000 BCE to 3000 CE and Rebellions in Qing

(a) cumulative distribution: value of coefficients



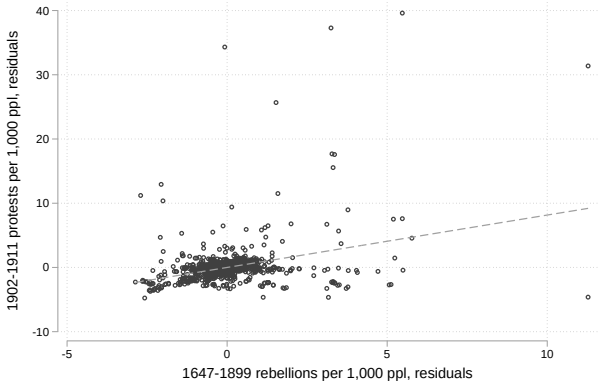

(b) coefficients and p-values



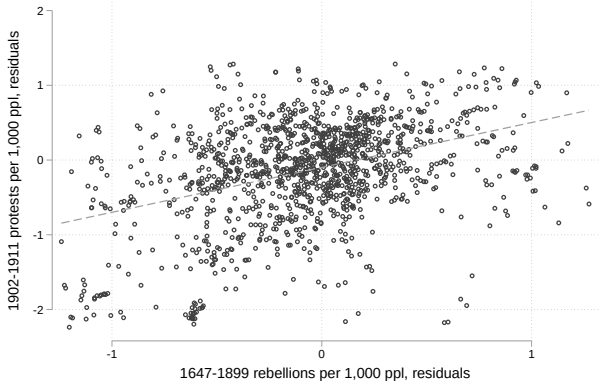

**Notes:** This figure reports the results of a placebo test described in section 5.1. We regress Qing dynasty rebellions on total eclipses observed in China during the Qing dynasty period as well as periods of equal length outside of the Qing dynasty period between 2000BCE and 3000CE. Panel (a) presents the cumulative distribution of the coefficients. Panel (b) presents a scatter plot of the coefficients and their corresponding p-values. These estimates use the same specification with Column 3 in Table 4, i.e. with Province FE, County Population Category FE, and full County controls. Standard errors clustered at the prefecture level.

## Figure 8: Correlation between 1911 Revolution protests and Qing rebellions per 1,000 people

(a) all counties




(b) exclude top-bottom 5%




**Notes:** This figure reports the correlation between total number of peasant uprisings observed in each county between 1647 and 1899 and total number of Xinhai-related rebellions observed in each county in the years 1902-1911. Both variables are standardized using population estimates from Skinner et al. (2011) and in per 1,000 people. The plotted points are residuals from regressing each variable on province fixed effects and county-level controls. County controls include log agricultural land tax per capita (1820), suitability of sweet potato, latitude degrees, longitude degrees, log distance to coast, and the number of eclipses from 2000BCE to 3000CE excluding the Qing dynasty.

# Table 1: Summary Statistics

| Variable | Obs | Mean | SD | Min | Max |
|---|---|---|---|---|---|
| **Panel A: Panel Variables** | | | | | |
| Rebellion dummy | 407650 | 0.005 | 0.069 | 0.000 | 1.000 |
| Eclipse dummy | 407650 | 0.003 | 0.054 | 0.000 | 1.000 |
| **Panel B: Cross-sectional Variables** | | | | | |
| Rebellions per 1,000 ppl (1647-1899) | 1550 | 0.622 | 1.255 | 0.000 | 14.000 |
| Num of eclipses (1647-1899) | 1550 | 0.758 | 0.895 | 0.000 | 4.000 |
| Xinhai-related rebellions per 1,000 ppl | 1550 | 0.539 | 2.693 | 0.000 | 44.000 |
| Legend dummy | 1550 | 0.297 | 0.457 | 0.000 | 1.000 |
| Temple dummy | 1550 | 0.276 | 0.447 | 0.000 | 1.000 |
| Martyr count in the county | 1550 | 0.615 | 1.394 | 0.000 | 14.000 |
| Importance level in governance | 1550 | 1.704 | 1.266 | 0.000 | 4.000 |

**Notes:** Panel A presents summary statistics for the main variables in panel regression. Panel B presents summary statistics for the main variables in cross-sectional regressions including sum of Qing and Xinhai rebellions (normalized by population and measured as the number per 1,000 people), sum of eclipses in Qing, and variables in the mechanism section. All variables are at the county level. See Section 3 and Table A.2 for detailed variable definitions and data sources.

# Table 2: Balance Test for Eclipses

| Panel A: Time-Varying Characteristics | (1)<br>1(eclipse) | (2)<br>1(eclipse) |
|---|---|---|
| Famine | 0.001 | 0.002 |
| | (0.001) | (0.002) |
| Flood | -0.002 | -0.000 |
| | (0.003) | (0.001) |
| Drought | 0.003 | 0.001 |
| | (0.003) | (0.001) |
| Insect infestation | 0.001 | 0.000 |
| | (0.002) | (0.002) |
| Earthquake | 0.001 | 0.002 |
| | (0.001) | (0.002) |
| New infrastructure | 0.001 | 0.004 |
| | (0.001) | (0.006) |
| Tax relief | 0.000 | 0.000 |
| | (0.002) | (0.001) |
| N | 407,650 | 407,650 |
| N eclipses | 1,177 | 1,177 |
| F statistic | | 0.473 |
| P value | | 0.854 |
| County FE | Yes | Yes |
| Province year FE | Yes | Yes |
| Panel B: Time-Invariant Characteristics | N Eclipses (Standardized) | N Eclipses (Standardized) |
| Log population (1893) | -0.016 | -0.217*** |
| | (0.032) | (0.076) |
| Log agricultural land tax in silver units per capita (1820) | -0.106** | -0.254*** |
| | (0.046) | (0.089) |
| Number of clans | -0.031 | 0.002 |
| | (0.034) | (0.034) |
| Crop suitability - fox millet | 0.008 | -0.000 |
| | (0.063) | (0.063) |
| Crop suitability - wet rice | 0.008 | -0.107* |
| | (0.055) | (0.056) |
| Crop suitability - sweet potato | 0.313*** | 0.313*** |
| | (0.077) | (0.085) |
| Latitude degrees | 0.681*** | 0.432* |
| | (0.234) | (0.240) |
| Longitude degrees | 0.321** | 0.228 |
| | (0.159) | (0.168) |
| Terrain ruggedness | 0.056 | 0.036 |
| | (0.046) | (0.046) |
| Log distance to Beijing | -0.168 | -0.049 |
| | (0.198) | (0.131) |
| Log distance to province Capital | 0.038 | 0.027 |
| | (0.028) | (0.022) |
| Log distance to the coastline | -0.155** | -0.165** |
| | (0.077) | (0.080) |
| Log distance to Yangtze River | 0.019 | 0.001 |
| | (0.058) | (0.054) |
| Number of eclipses (2000 BCE – 3000 CE, excluding Qing) | -0.095* | -0.094* |
| | (0.051) | (0.056) |
| N | 1,550 | 1,550 |
| N eclipse counties | 806 | 806 |
| F statistic | | 4.051 |
| P value | | 0.000 |
| Province FE | Yes | Yes |

**Notes:** Column 1 reports point estimates from regressions of time-varying (Panel A) and time-invariant (Panel B) county characteristics on a dummy variable indicating the occurrence of a total eclipse. Column 2 reports point estimates from regressions of eclipse count (in standard deviations) on all county characteristics in a single regression. Eclipse counties are those inside the totality zone of at least one solar eclipse during the Qing dynasty period. The reported F-Stat and p-value are calculated from the joint regression in Column 2.

## Table 3: The Effect of Total Eclipses on The Probability of Rebellions - Panel Specification

| | 1(rebellion) | | | | | |
|---|---|---|---|---|---|---|
| | | | | confucian temple | | |
| | | | | yes | no | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| 1(eclipse) | 0.126*** | 0.128*** | 0.117*** | 0.156*** | 0.066*** | 0.090*** |
| | (0.033) | (0.032) | (0.028) | (0.039) | (0.019) | (0.026) |
| 1(eclipse) × confucian temple | | | | | | 0.059** |
| | | | | | | (0.030) |
| Observations | 407,650 | 407,650 | 407,650 | 212,241 | 195,409 | 407,650 |
| R-squared | 0.049 | 0.204 | 0.222 | 0.262 | 0.228 | 0.223 |
| Year FE | y | n | n | n | n | n |
| County FE | y | y | y | y | y | y |
| Province × year FE | n | y | y | y | y | y |
| County controls | n | n | y | y | y | y |
| N clusters (prefectures) | 249 | 249 | 249 | 185 | 195 | 249 |

**Notes:** The table shows coefficient estimates corresponding to equation (1). The dependent variable 1(rebellion) is a dummy equal to 1 for county-year observations with a recorded rebellion in local chronicles and 0 otherwise. 1(eclipse) is a dummy variable equal to 1 for county-year observations within the totality zone of an eclipse in a given year, and 0 otherwise. County controls include all time varying county characteristics reported in Panel A of Table 2, log agricultural land tax per capita (1820), suitability of sweet potato, latitude degrees, longitude degrees, and log distance to coast interacted with year fixed effects. Standard errors clustered at the prefecture level reported in parenthesis. Significance level: *** p<0.01, ** p<0.05, * p<0.1.

## Table 4: The Effect of Past Eclipses on Incidence of Rebellions during Qing period

| | N Rebellions per 1,000 ppl (1647–1899) | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| (N Total Eclipses)$_{1647-1899}$ | 0.275*** | 0.250*** | 0.255*** |
| | (0.083) | (0.063) | (0.055) |
| Observations | 1,550 | 1,550 | 1,550 |
| R-squared | 0.039 | 0.152 | 0.401 |
| Province FE | n | y | y |
| County pop categories FE | n | n | y |
| County controls | n | n | y |
| N clusters (prefectures) | 249 | 249 | 249 |

**Notes:** The table shows coefficient estimates corresponding to equation (2). The dependent variable is the sum of rebellions per 1,000 people recorded between 1647 and 1899 in each county. The independent variable is the number of total eclipses observed in the county between 1647 and 1899. Variables in per 1,000 people are constructed using county population estimates from Skinner et al. (2011). County controls include log agricultural land tax per capita (1820), suitability of sweet potato, latitude degrees, longitude degrees, log distance to coast, and the number of eclipses from 2000BCE to 3000CE excluding the Qing dynasty. Standard errors clustered at the prefecture level reported in parenthesis. Significance level: *** p<0.01, ** p<0.05, * p<0.1.

# Table 5: The Effect of Past Rebellions on 1902-1911 Protests

| | N Xinhai Protests per 1,000 ppl (1902–1911) | | | | | |
|---|---|---|---|---|---|---|
| | OLS | | Reduced Form | | 2SLS | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| (N Rebellions per 1,000 ppl)$_{1647-1899}$ | 0.981*** | 0.816*** | | | 1.057** | 1.025** |
| | (0.183) | (0.188) | | | (0.441) | (0.470) |
| (N Total Eclipses)$_{1647-1899}$ | | | 0.264** | 0.261** | | |
| | | | (0.122) | (0.131) | | |
| Observations | 1,550 | 1,550 | 1,550 | 1,550 | 1,550 | 1,550 |
| R-squared | 0.215 | 0.239 | 0.039 | 0.155 | 0.187 | 0.106 |
| Province FE | y | y | y | y | y | y |
| County pop categories FE | n | y | n | y | n | y |
| County controls | n | y | n | y | n | y |
| N clusters (prefectures) | 249 | 249 | 249 | 249 | 249 | 249 |

**Notes:** The table shows coefficient estimates of OLS, reduced form and 2SLS specifications described in section 4.2. Columns (5) and (6) correspond to equation (3). The dependent variable is the sum of Xinhai-related protests per 1,000 people recorded between 1902 and 1911 in each county. The independent variable is the total number of rebellions per 1,000 people recorded between 1647 and 1899 in each county. The instrument (N Total Eclipses) is the number of total eclipses observed in the county between 1647 and 1899. Variables in per 1,000 people are constructed using county population estimates from Skinner et al. (2011). County controls include log agricultural land tax per capita (1820), suitability of sweet potato, latitude degrees, longitude degrees, log distance to coast, and the number of eclipses from 2000BCE to 3000CE excluding the Qing dynasty. Standard errors clustered at the prefecture level reported in parenthesis. Significance level: *** p<0.01, ** p<0.05, * p<0.1.

# Table 6: Potential Mechanisms of Persistence

**Panel A: Collective Memory - 2SLS**

| | 1(Legend) | | N(Martyrs) | | 1(Temple) | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| (N Rebellions per 1,000 ppl)$_{1647-1899}$ | 0.127** | 0.141** | 0.335* | 0.333* | 0.192* | 0.195** |
| | (0.061) | (0.065) | (0.181) | (0.172) | (0.099) | (0.085) |
| Observations | 1,550 | 1,550 | 1,550 | 1,550 | 1,550 | 1,550 |
| Province FE | y | y | y | y | y | y |
| County pop categories FE | y | y | y | y | y | y |
| County controls | n | y | n | y | n | y |
| N clusters (prefectures) | 249 | 249 | 249 | 249 | 249 | 249 |
| Depvar mean | 0.297 | 0.297 | 0.615 | 0.615 | 0.276 | 0.276 |

**Panel B: Governance - 2SLS**

| | Importance Level (0-4) | Chong Important in Trans-portation | Fan Complicated in Public Affairs | Pi Difficult to Collect Tax | Nan High in Crime |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| (N Rebellions per 1,000 ppl)$_{1718-1747}$ | 2.035** | 0.109 | 0.595 | 0.760* | 0.572** |
| | (0.941) | (0.266) | (0.386) | (0.422) | (0.246) |
| Observations | 1,550 | 1,550 | 1,550 | 1,550 | 1,550 |
| Province FE | y | y | y | y | y |
| County pop categories FE | y | y | y | y | y |
| County controls | y | y | y | y | y |
| N clusters (prefectures) | 249 | 249 | 249 | 249 | 249 |
| Depvar mean | 1.704 | 0.394 | 0.543 | 0.246 | 0.521 |

**Panel C: Social capital - 2SLS**

| | 1(Clan) | | N(Clans) | | Clans per 1,000 ppl | |
|---|---|---|---|---|---|---|
| VARIABLES | (1) | (2) | (3) | (4) | (5) | (6) |
| (N Rebellions per 1,000 ppl)$_{1647-1899}$ | 0.020 | 0.019 | -2.573 | 0.327 | -2.096 | -1.053 |
| | (0.059) | (0.057) | (3.075) | (3.193) | (1.307) | (1.180) |
| Observations | 1,550 | 1,550 | 1,550 | 1,550 | 1,550 | 1,550 |
| Province FE | y | y | y | y | y | y |
| County pop categories FE | y | y | y | y | y | y |
| County controls | n | y | n | y | n | y |
| N clusters (prefectures) | 249 | 249 | 249 | 249 | 249 | 249 |
| Depvar mean | 0.373 | 0.373 | 7.350 | 7.350 | 2.531 | 2.531 |

**Notes:** In Panel A, Columns 1, 2, 5, and 6 represent dummy variables indicating whether each county has records of martyrs celebrated in local legends or temples; Columns 3 and 4 represent the number of important anti-Qing martyrs recorded in each county chronicles. In Panel B, Column 1 represents the cumulative importance level for each county, which is the sum up of four separate indicators in Column 2-5. In Panel C, Column 1 and 2 represent dummy variable indicating whether each county has records of clans. Column 3 and 4 represent the number of clans in each county. Column 5 and 6 represent clans per 1,000 people scaled by population in Qing. County controls include log agricultural land tax per capita (1820), suitability of sweet potato, latitude degrees, longitude degrees, log distance to coast, and the probability of eclipses from 2000BCE to 3000CE excluding the Qing dynasty. Standard errors clustered at the prefecture level reported in brackets. Significance level: *** p<0.01, ** p<0.05, * p<0.1.

# Table 7: The Effect of Past Rebellions on Current Protests

**Panel A: CASS Data - County Level 2SLS**

| | All Protests | | Target: Local Government | | Target: Non-Local Government | |
|---|---|---|---|---|---|---|
| | Protest per 1M ppl | 1(Protest) | Protest per 1M ppl | 1(Protest) | Protest per 1M ppl | 1(Protest) |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| (N Rebellions per 1,000 ppl)$_{1647-1899}$ | 0.036* | 0.116** | 0.029 | 0.081* | 0.008 | 0.034 |
| | (0.022) | (0.054) | (0.018) | (0.042) | ( 0.012) | (0.039) |
| Observations | 1,550 | 1,550 | 1,550 | 1,550 | 1,550 | 1,550 |
| Province FE | y | y | y | y | y | y |
| County controls | y | y | y | y | y | y |

**Panel B: CASS and GDELT Data - Prefecture Level 2SLS**

| | CASS | | GDELT | |
|---|---|---|---|---|
| | Protests per 1M ppl | 1(Protest) | Protests per 1M ppl | 1(Protest) |
| | (1) | (2) | (3) | (4) |
| (N Rebellions per 1,000 ppl)$_{1647-1899}$ | 0.116 | 1.170* | 0.648 | 0.702* |
| | (0.071) | (0.674) | (1.824) | (0.348) |
| Observations | 270 | 270 | 270 | 270 |
| Province FE | y | y | y | y |
| Prefecture controls | y | y | y | y |

**Panel C: CLB and ACLED Data - Prefecture Level 2SLS**

| | CLB | | ACLED | |
|---|---|---|---|---|
| | Protests per 1M ppl | 1(Protest) | Protests per 1M ppl | 1(Protest) |
| | (1) | (2) | (3) | (4) |
| (N Rebellions per 1,000 ppl)$_{1647-1899}$ | 1.976* | 0.060 | 1.762 | 0.212* |
| | (1.117) | (0.156) | (3.144) | (0.118) |
| Observations | 270 | 270 | 270 | 270 |
| Province FE | y | y | y | y |
| Prefecture controls | y | y | y | y |

**Notes:** The dependent variable in odd columns is the average number of social unrest episodes each year per 1 million people, standardized by population in 2000. The dependent variable in even columns is a dummy indicating whether social unrest episodes occurred. Panel A presents the test results using the CASS data at the county level. This dataset allows us to distinguish protests targeting local governments from those targeting non-local governments. It only recorded protests with more than 100 participants. For comparison, we also collapse CASS data to the prefecture level. Panel B presents the test results using the prefecture-level CASS and GDELT data. Panel C presents the test results using the prefecture-level CLB (with more than 100 participants) and ACLED data. County controls include log agricultural land tax per capita (1820), suitability of sweet potato, latitude degrees, longitude degrees, log distance to coast, and the probability of eclipses from 2000 BCE to 3000 CE excluding the Qing dynasty. Standard errors clustered at the prefecture level are reported in brackets. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

# Appendix

This Appendix provides supplementary figures, tables, and additional details referenced in the main text.

# A Details on Data Sources and Construction Process of Key Historical Variables

## 1. Martyrs' Legends

Our data on martyrs come from the Emperor Designated Book of Pre-dynasty's Martyred Officials (Wuying Hall Scholars, 1776), an official Qing compilation documenting those who died resisting the Qing conquest during the Ming–Qing transition. The conquest of the Southern Ming regimes involved prolonged warfare in which many Ming officials and local elites chose to die rather than surrender. Their sacrifices generated extensive local records and family accounts commemorating loyalist martyrdom.

The Qianlong Emperor ordered the compilation of the *Qinding Shengchao Xunjie Zhuchen Lu* as part of a broader effort to consolidate the Qing court's historical narrative and political legitimacy. The project aimed to collect, verify, and canonize officials who had died in loyalty to the Ming, producing a standardized, court-sanctioned register of martyrs. A companion volume, the *Erchen Zhuan*, documented those deemed turncoats.

The work comprises 12 volumes and records 3,787 individuals (including family members who died together), primarily Southern Ming central and local officials, along with some military commanders and prominent gentry. Each entry provides the individual's native place, official position, date of death, and circumstances of martyrdom. Following its compilation, the Qing state conferred posthumous honors on many of these individuals, formally incorporating them into state-recognized commemorative institutions.

We restrict our sample to the 26 major individuals in the first volume, whose entries are the most detailed and who held the greatest historical prominence. These figures appear frequently in local chronicles, which enables us to measure how their stories were transmitted at the local level. To construct our heroic-portrayal measure, we searched local chronicles for all sentences mentioning each individual by name. We did data cleaning for our querying results for two reasons: (1) Since these martyrs mostly held important official positions and were quite prestigious, their actions, even those unrelated to the resistance, were recorded in

the local chronicles[26]. (2) A few names are only with two Chinese characters. Thus, a simple search would capture references to unrelated individuals[27].

To ensure we identify genuine references to the martyrs, we constructed a list of 13 keywords related to bravery, resistance, and typical good character. We then narrow our records containing these keywords as evidence that people in the Qing period commemorate these officials and pass on their spirit of resistance as a memory. These keywords are: bravery (勇), martyrdom (烈), fight (斗), sacrifice (捐), integrity or no corruption (廉), filial piety (孝), good reputation (名), resistance (抗), heroism or generosity (豪), backbone (骨)[28], sympathy (哀), loyalty (忠), and heroic (英). Our baseline measure uses 10 keywords jointly in group 4 of Table A.8, which yields 2,515 textual references describing the heroic actions of anti-Qing loyalists.

To show that the results do not depend on the specific set of keywords used to identify martyrs, Table A.8 also reports 2SLS estimates obtained by progressively expanding the keyword list from a seven-keyword combination to all 13 keywords. Group 1 uses only the seven core terms most directly related to resistance and official virtue (bravery, fight, martyrdom, sacrifice, resist, integrity or no corruption, filial piety), while subsequent groups add one keyword at a time. As shown, the 2SLS coefficients are stable across all keyword combinations, both when using the legend dummy and the count of martyrs as the independent variable. The number of counties with martyrs' records increases as more keywords are included, from 356 in Group 1 to 564 in Group 7. The point estimates remain similar in magnitude throughout the process, indicating that the results are not driven by the choice of any particular keyword.

## 2. Temples

Our temple data are drawn from local chronicles compiled at the prefectural and county levels. These chronicles typically record temples in dedicated sections, providing information on their names, locations, and purposes. Research assistants and graduate students in the History Department read all temple entries and identified those dedicated to loyal martyrs or otherwise commemorating martial valor and resistance to the Qing conquest. We excluded temples whose recorded construction date fell outside the Qing dynasty.

We then geocoded the temple records by matching the place names in the chronicles to the

[26]For example, the Renqiu County Chronicle from the Qianlong Emperor recorded a memorial article for a local centenarian written by one of the martyrs, Sun Chengzong, who was a high-ranking official. This record has no connection to people commemorating his heroic anti-Qing actions.

[27]For example, Liu Ting is a simple name, and we will find records of other people named Liu Ting.

[28]This word originally means "bone" in Chinese. However, it can also mean "fallen soldiers" in ancient Chinese poems.

CHGIS database (CHGIS, 2016). When a chronicle entry specified only the prefecture, we assigned the temple to the prefecture's capital county (e.g., temples listed under Guangzhou Prefecture were mapped to Guangzhou County). When only the province was specified, we assigned the entry to the central county of the province's capital prefecture. This procedure yields 720 temple records distributed across 428 counties.

## 3. Counties' Importance Rankings

The Qing dynasty classified counties according to four binary attributes—*chong* (冲), *fan* (繁), *pi* (疲), and *nan* (难)—which together captured the administrative difficulty of each post. *Chong* denoted importance in transportation; *fan*, a place with numerous and complex administrative affairs; *pi*, a county frequently in arrears on taxes and grain payments; and *nan*, an area with an unruly population or frequent crime. Each attribute functions as a dummy variable, so a county could hold between zero and four attributes.

The number of attributes determined how officials were appointed. Counties with zero, one, or two attributes were considered routine posts, and vacancies were filled by the Ministry of Personnel (*Libu*) through a lottery-based system. Counties with three or four attributes were designated important posts, with appointments made directly by the emperor and provincial governors-general.

The system was introduced in 1728 by Jin Hong, Provincial Administrative Commissioner of Guangxi, and endorsed by the Yongzheng Emperor. It underwent nationwide revisions in 1733, 1741, and 1747. The 1733 review adjusted county classifications to better reflect local conditions. The 1741 reform, initiated by the Qianlong Emperor, imposed a more systematic correspondence between the number of attributes and the rank and appointment authority of posts. The 1747 revision reduced the number of attributes assigned to counties and shifted more appointments back to the Ministry of Personnel, aiming to curb frequent transfers and stabilize local governance. No further revisions of comparable scale were undertaken thereafter.

Our data record each county's *chong–fan–pi–nan* classification as of 1893 and are from the CHGIS China-W data.

## 4. Clan Records

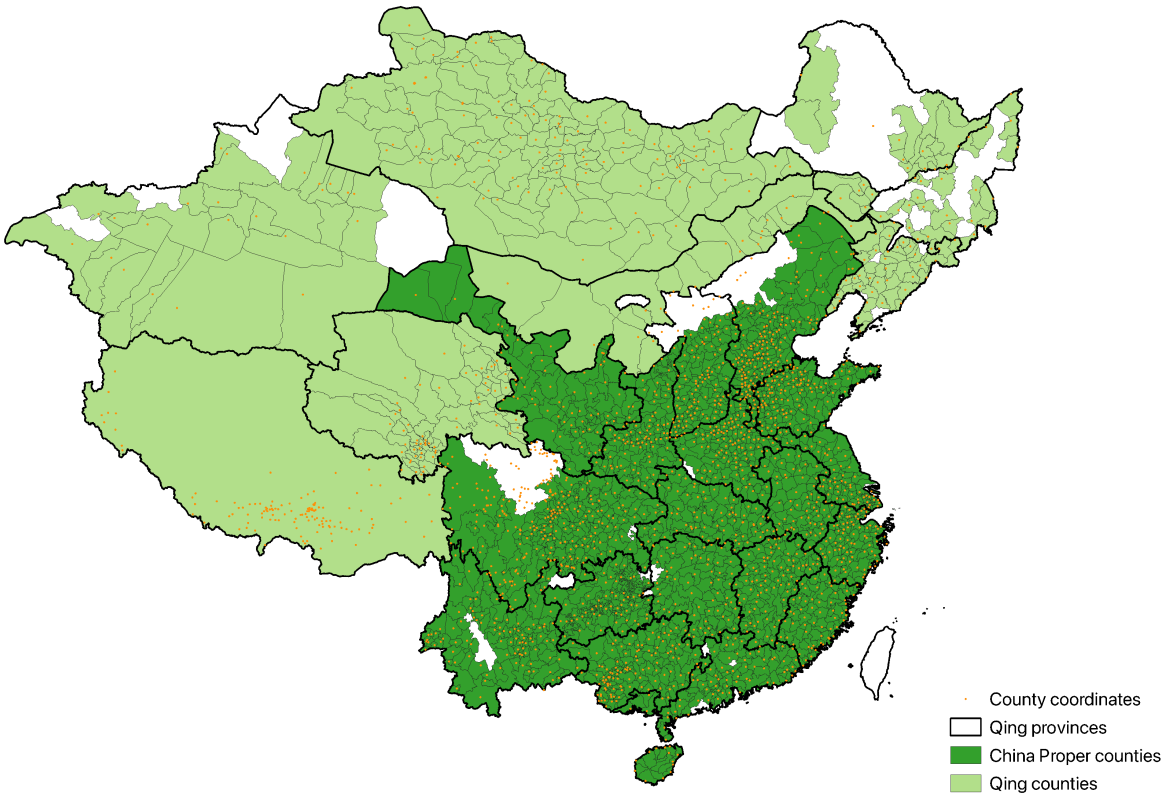
Our clan data come from the Chinese Genealogy Knowledge Service Platform organized by Shanghai Library, the largest repository of original Chinese clan records, holding approximately 40,000 titles and over 380,000 volumes. The collection spans more than 20 provinces, with particularly strong coverage in Zhejiang, Hunan, Jiangsu, and Anhui. Many of the

records are woodblock or movable-type printed editions compiled during the Qing dynasty and the Republican period.

From this database, we extract each clan's surname and registered place of origin. Most records were compiled during the Qing or the Republican period (1912–1949). Because clan structures were highly persistent, Republican-era records tend to capture clan networks that were already well established during the Qing, and thus remain informative about earlier clan presence and local prominence.

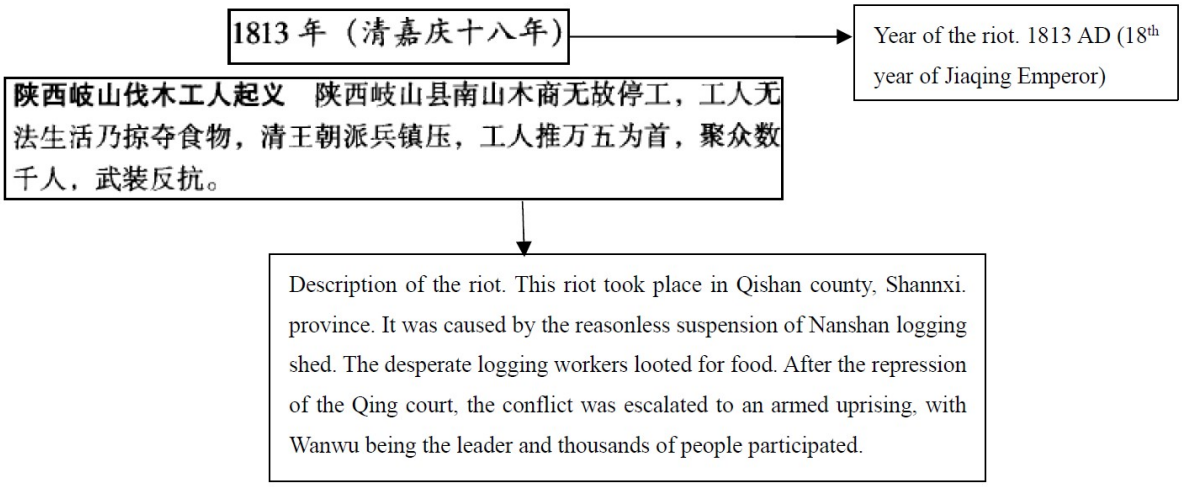
# Additional Tables and Figures

## Figure A.1: Qing counties and provinces in CHGIS V6



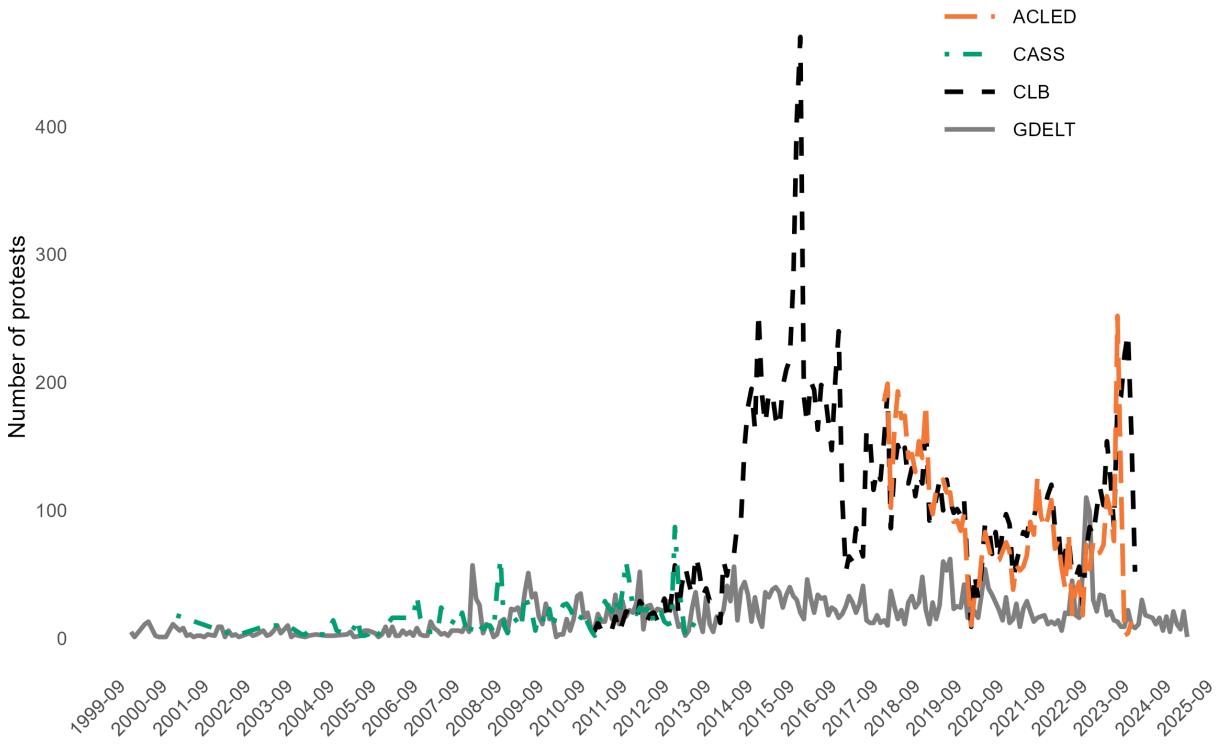

This figure plots all counties in CHGIS V6. Counties in China Proper are plotted in dark green. Orange dots are counties' coordinates as reported CHGIS V6. Province boundaries are plotted in thick black lines. In blank we report areas where geographical information is missing in the original shapefiles.

## Figure A.2: Example of Local Protest Record



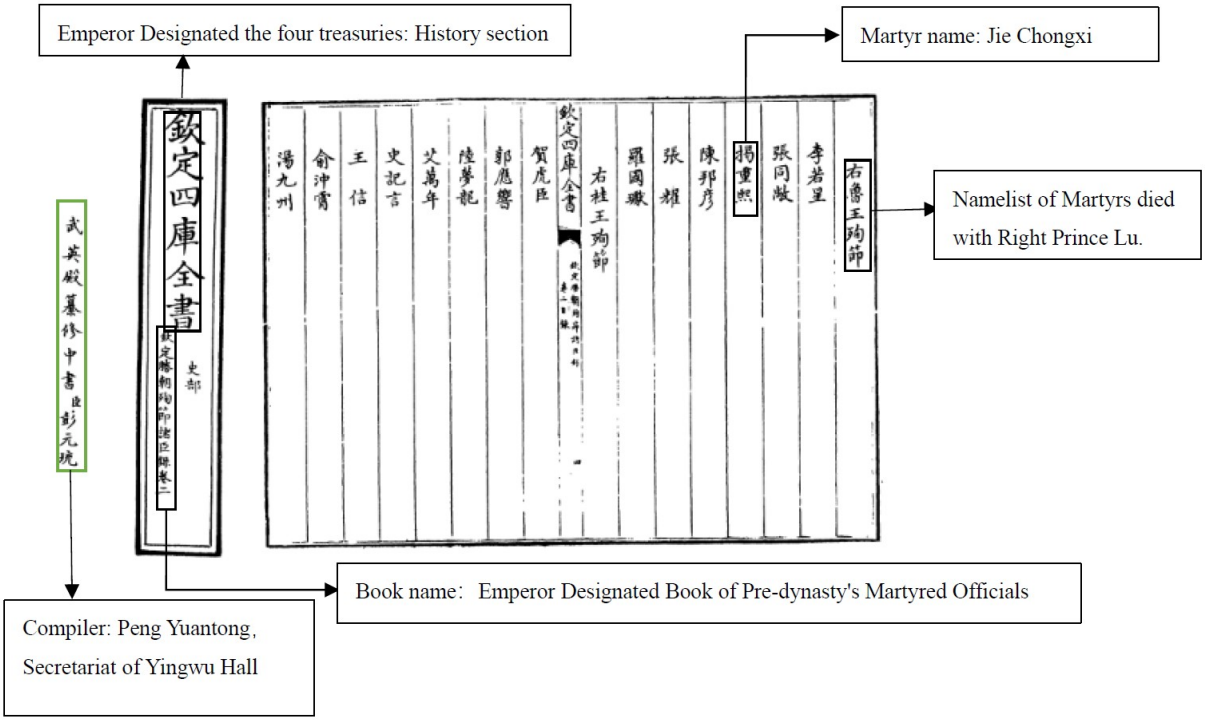

**Notes:** This figure shows the record of a local rebellion as it appears in the book "Chronology of Wars in Chinese History" (Zhongguo Lidai Zhanzheng Nianbiao). The year of the riot and a short description regarding the location, the reason, the leader and number of participants is provided. It also provides a short discussion of the rebellion and the government's reaction.

## Figure A.3: Comparing the social unrest records of CASS, ACLED, CLB, and GDELT



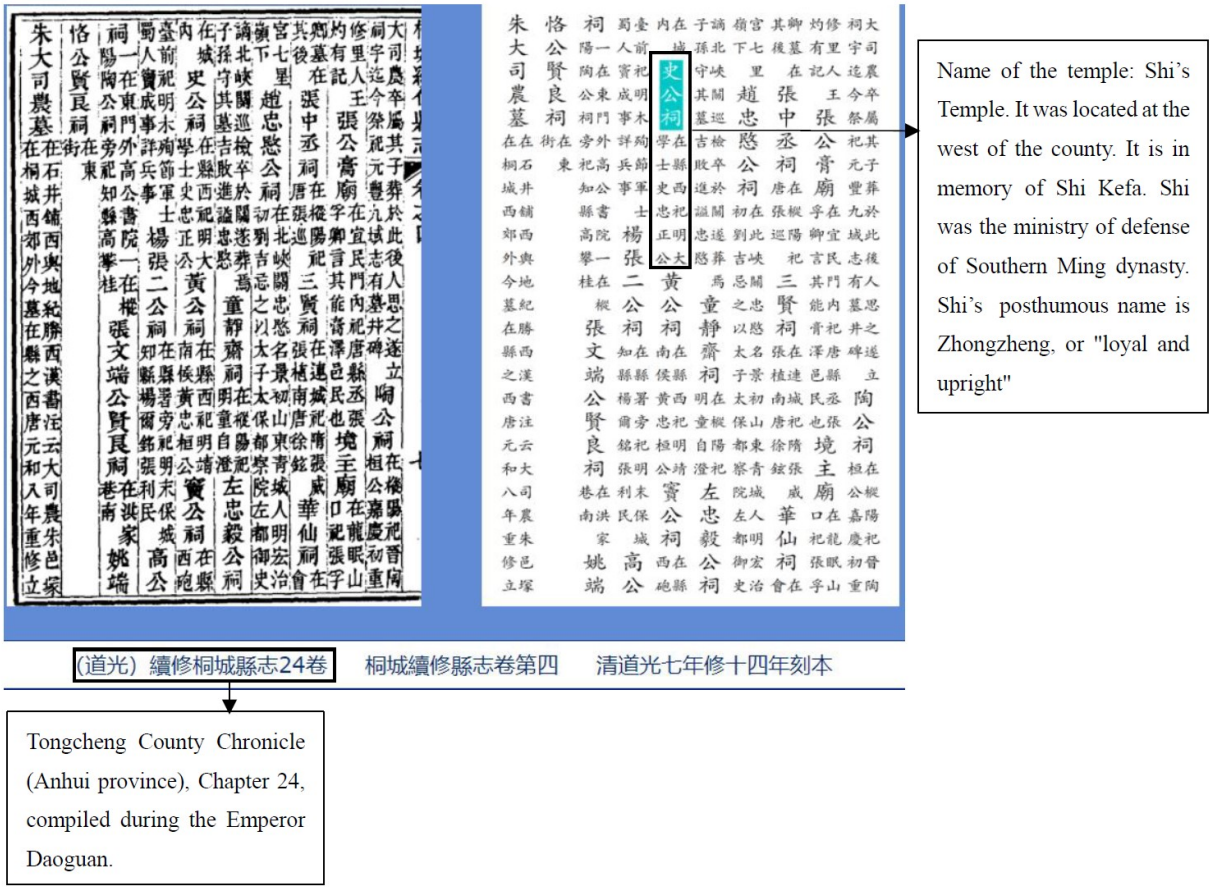

**Notes:** The figure shows the number of unrest episodes by month reported in CASS, ACLED, CLB, and GDELT. These are the social unrest events used in Table 7.

# Figure A.4: Example of anti-Qing Martyr Record



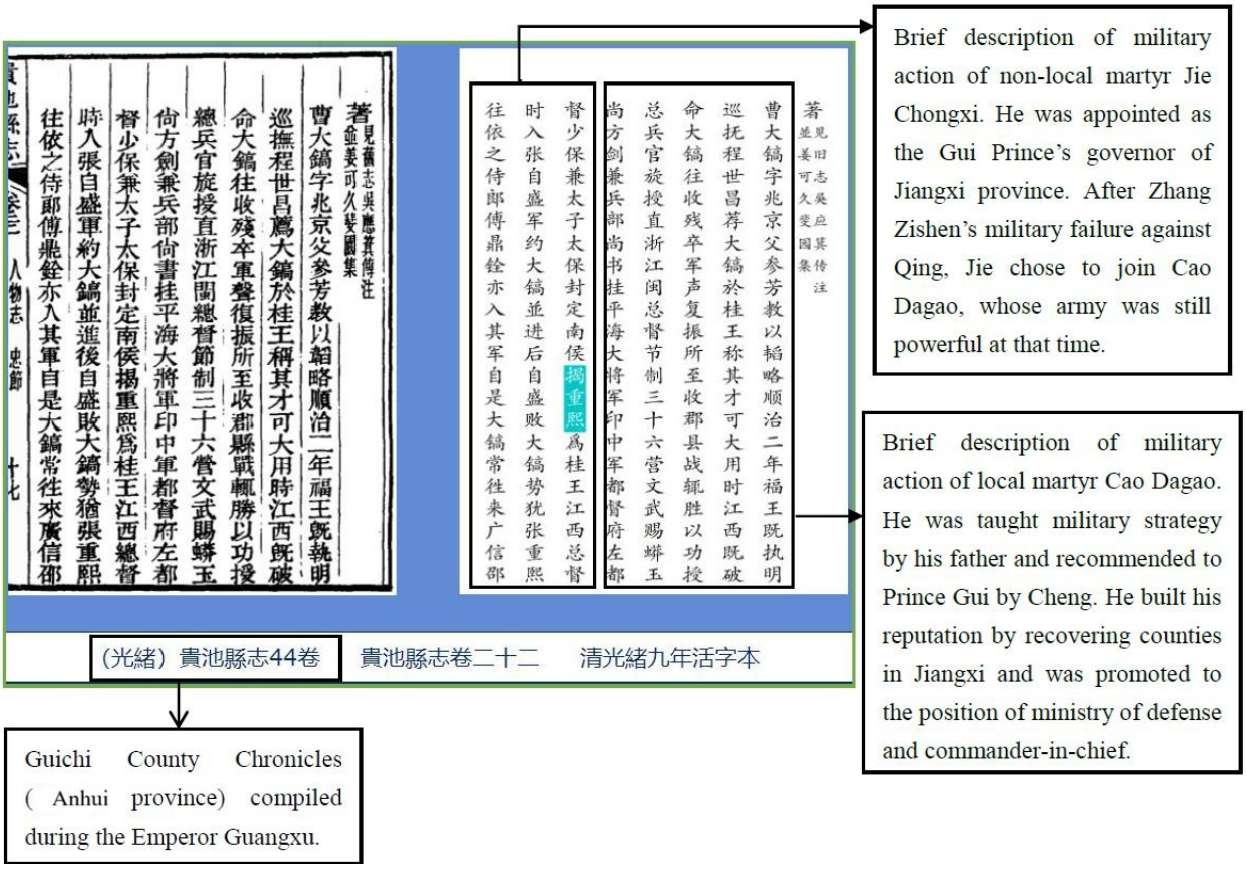

**Notes:** The figure shows an example of anti-Qing martyr extracted from the "Emperor Designated Book of Pre-dynasty's Martyred Officials". The book is part of the History Section of the Siku Quanshu (translated as the Complete Library in Four Sections, or the Emperor Designated of the four Treasuries), a massive collection of books compiled by the Qing court at the end of the 18th century. The names of the martyrs are ordered according to the princes they served and died with.

# Figure A.5: Example of anti-Qing Martyr's Temple Record




**Notes:** This figure shows an example of the location of the martyrs' temples. The temple location information comes from the local chronicles. It also provides the posthumous title of the martyr.

## Figure A.6: Example of anti-Qing Martyr's Legend Record




**Notes:** This figure shows an example of the martyrs' legends. On top of the martyr who was originated from the county, e.g., Cao Dagao, it also mentioned his non-local associated and also a martyr, Jie Chongxi.

## Table A.1: Information about different current protests data sources

| Data | Year | N(records) | Geo info | Analysis level | Geo-located protests |
|---|---|---|---|---|---|
| CASS | 2001-2013 | 1,048 | County name | County & Prefecture | 1,048 |
| ACLED | 2018-2023 | 10,625 | Coordinates | Prefecture | 6,445 |
| CLB | 2011-2023 | 16,954 | Prefecture name | Prefecture | 15,799 |
| GDELT | 2000-2023 | 6,065 | Coordinates | Prefecture | 5,082 |

**Notes:** This table presents the basic information for different current protests data sources. The CASS dataset only records protests with more than 100 participants. Most data in the CLB is recorded on a daily basis, meaning that if a protest occurs on two consecutive days for the same event, it will be recorded as two records in the CLB. The ACLED dataset largely comes from the CLB. The raw CLB data only focuses on mainland China, while ACLED also records data from Hong Kong and Macau. However, since our main analysis excluded Hong Kong and Macau, the two data show similar trends for the overlapping time period in Figure A.3.

## Table A.2: Variable Definitions and Data Sources

| Variable Description | Data Source | Level of Variation |
|---|---|---|
| Coordinate-based and polygon-based county ID | China Historical Geographic Information System Version 6 (CHGIS V6) (CHGIS, 2016), access at `https://chgis.fas.harvard.edu/data/chgis/v6/`.[29] | County |
| Latitude of county centroid | | County |
| Longitude of county centroid | | County |
| Dummy for whether a total eclipse's path crossed the county point in a specific year | NASA: Five Millennium Catalog of Solar Eclipses (Espenak and Meeus (2006a)) | County |
| Total number of historical eclipses (2000 BCE–3000 CE) affecting the county, excluding the Qing period | | County |
| Cumulative number of eclipses affecting the county, 1647–1899 | | County |
| Cumulative number of eclipses affecting the county across 21 non-overlapping historical periods spanning 2000 BCE–3000 CE | | County |
| Dummy for whether a rebellion occurred in the county-year | Book: Chronology of Wars in Chinese History (The Chinese Military History Editorial Team (2003)). Institutional access needed. Manually Collected. | County |
| Cumulative number of rebellions, 1647–1899, per 1,000 population | | County |
| Dummy for whether the county had a Confucius temple | Erudition Chinese Local Chronicles Database (Beijing Erudition Digital Research Center, 2025)[30]. By searching keywords. | County |
| Count or dummy for whether the county has a recorded legend of anti-Qing martyrs | Emperor Designated Book of Pre-dynasty's Martyred Officials (Wuying Hall Scholars, 1776) and Erudition Chinese Local Chronicles Database (Beijing Erudition Digital Research Center, 2025). | County |
| Number of anti-Qing martyrs in the county | | County |

[29]Full citation: CHGIS, Version: 6. (c) Fairbank Center for Chinese Studies of Harvard University and the Center for Historical Geographical Studies at Fudan University, 2016. In the following descriptions, it will be abbreviated as CHGIS V6. Distribution is not permitted.

[30]This is a commercial database of digitalized ancient books. Institutions are required to purchase at least volumes 1-4. For details, please refer to: `http://er07.com/home/pro_87.html`.

| Variable Description | Data Source | Level of Variation |
|---|---|---|
| Dummy for whether taxes collected in the county were lower than the fixed quota set by the Qing government | Routine Memorials made by six major departments of the central government in the Qing Dynasty to the emperor (First Historical Archives of China, 2025). Institutional access needed. Manually Collected. | County |
| Cumulative count of years for the dummy of lower taxation | | County |
| Dummy for new-infrastructure-related activity in the county | | County |
| Cumulative count of new infrastructure related activity | | County |
| Dummy for whether the county experienced a natural disaster | Erudition Chinese Local Chronicles Database (Beijing Erudition Digital Research Center, 2025) | County |
| Dummy for whether the county experienced a famine | | County |
| Dummy for whether the county experienced a flood | | County |
| Dummy for whether the county experienced a drought | | County |
| Dummy for whether the county experienced an insect plague | | County |
| Dummy for whether the county experienced an earthquake | | County |
| Count or dummy for whether the county had a temple commemorating martyrs | | County |
| Log distance from the county to Beijing | CHGIS V6 (CHGIS, 2016) | County |
| Log distance from the county to its provincial capital | | County |
| Log distance from the county to the coastline | OSMCoast500 V1 (OpenStreetMap Contributors (2024)), accessed at `https://data.csiro.au/collection/csiro:57386` in September, 2025 and CHGIS V6 (CHGIS, 2016) | County |
| Log distance from the county to the Yangtze River system | *Natural Earth* Rivers + lake centerlines (Natural Earth (2022)), accessed at `https://www.naturalearthdata.com/downloads/` in September, 2025 and CHGIS V6 (CHGIS, 2016) | County |

| Variable Description | Data Source | Level of Variation |
|---|---|---|
| Foxtail millet crop suitability index | FAO GAEZ v4 Section 4 (Food and Agriculture Organization of the United Nations and International Institute for Applied Systems Analysis (2021)), accessed at `https://gaez.fao.org/pages/theme-details-theme-4` in August, 2025 and CHGIS V6 (CHGIS, 2016) | County |
| Rice crop suitability index | | County |
| Sweet potato crop suitability index | | County |
| Number of clans in the county | Chinese Genealogy Knowledge Service Platform (Shanghai Library, 2025). Open access: `https://jiapu.library.sh.cn/` | County |
| Dummy of clan existence | | County |
| Number of clans in the county per 1,000 population | | County |
| Log agricultural land tax in silver units per capita (1820) | Historical Statistics of Population, Land and Prefecture. Obtained from Bai and Jia (2016). | Prefecture |
| Qing Dynasty Population in 1893 | ChinaW – Cities, County Seats and Yamen Units (1820–1893) Skinner et al. (2011). Accessed at `https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/JCT5NE` | County |
| Qing Dynasty Population in 1900 | Hu (1935). Published online: `https://www.geog.com.cn/CN/10.11821/xb193502002`. | County |
| Chinese county population in 2000 | National Bureau of Statistics of China (2001). Institutional access needed. | |
| Importance level component: strategically located for transportation | CHGIS V6 (CHGIS, 2016) – ChinaW (Skinner et al. (2011)), accessed at `https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/JCT5NE`. Level in 1893. | County |
| Importance level component: administratively complex in public affairs | | County |
| Importance level component: problematic tax collection | | |
| Importance level component: difficult due to violent customs and prone to crime | | County |

| Variable Description | Data Source | Level of Variation |
|---|---|---|
| Overall importance level constructed from all four above components | | County |
| Rebellions during the Xinhai Revolution, per 1,000 population | The Chronology of Civil Protests in the Late Qing Dynasty (Ding and Zhang (1982)). Obtained from Hao et al. (2022). | County |
| Total number of protests in the county, per 1,000,000 population | Institute of Law, Chinese Academy of Social Sciences (CASS) (Institute of Law, Chinese Academy of Social Sciences (CASS) (2023)). Distribution of CASS data is not permitted. People need institutional access with Institute of Law, Chinese Academy of Social Sciences | County |
| Number of protests targeting local government, per 1,000,000 population | | County |
| Number of protests targeting non-local government, per 1,000,000 population | | County |
| Number of protest events recorded by GDELT, per 1,000,000 population | Global Database of Events, Language, and Tone (GDELT) (The GDELT Project (2013)), accessed at `https://www.gdeltproject.org/data.html` on April 9, 2025. | Prefecture |
| Number of strikes/protests recorded by CLB, per 1,000,000 population | China Labour Bulletin (2023)[31] | Prefecture |
| Number of protest events recorded by ACLED, per 1,000,000 population | Armed Conflict Location & Event Data Project (2024), accessed at `https://acleddata.com` on January 13, 2024. Distribution of ACLED data is not permitted. | Prefecture |
| Terrain ruggedness index | Global Mapping International's Seamless Digital Chart of the World Base Map. Open access at `https://www.worldgeodatasets.com/basemaps/`; Method: Nunn and Puga (2012) | County |
| Prefecture-level administrative code used to merge contemporary protest datasets (CLB, GDELT, ACLED) | UN OCHA Common Operational Datasets (United Nations Office for the Coordination of Humanitarian Affairs (2020)). Open access at `https://data.humdata.org/dataset/cod-ab-chn`. Status in 2020. | Prefecture |

[31]Previously publicly accessible via the CLB Strike Map at `https://maps.clb.org.hk/strikes/en`. We accessed the data on January 13, 2024. However, CLB has ceased operations since 2025, so the live site is no longer maintained and cannot be accessed.

# Table A.3: The Effect of Total Eclipses on The Probability of Rebellions Additional Results

| | 1(rebellion) | | | | | | |
|---|---|---|---|---|---|---|---|
| | baseline | PPML | excl. Taiping (excl. 1850-1864) | early period pre-1800 | late period 1800-1909 | subsequent shocks | eclipse during natural disasters |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| 1(eclipse) | 0.117*** | 3.616*** | 0.118*** | 0.179*** | 0.037* | 0.101*** | 0.115*** |
| | (0.028) | (0.390) | (0.028) | (0.043) | (0.022) | (0.024) | (0.028) |
| 1(eclipse) × 1(past eclipse) | | | | | | 0.057* | |
| | | | | | | (0.033) | |
| 1(past eclipse) | | | | | | 0.002 | |
| | | | | | | (0.001) | |
| 1(eclipse) × 1(natural disaster) | | | | | | | 0.042 |
| | | | | | | | (0.052) |
| 1(natural disaster) | | | | | | | 0.000 |
| | | | | | | | (0.000) |
| Observations | 407,650 | 94,560 | 384,400 | 237,150 | 170,500 | 407,650 | 407,650 |
| R-squared | 0.222 | | 0.150 | 0.187 | 0.242 | 0.223 | 0.222 |
| Year FE | n | y | n | n | n | n | n |
| County FE | y | y | y | y | y | y | y |
| Province × year FE | y | n | y | y | y | y | y |
| County controls | y | y | y | y | y | y | y |
| N clusters (prefectures) | 249 | 160 | 249 | 249 | 249 | 249 | 249 |

**Notes:** The dependent variable 1(rebellion) is a dummy equal to 1 for county-year observations with a recorded rebellion in local chronicles and 0 otherwise. 1(eclipse) is a dummy variable equal to 1 for county-year observations within the totality zone of an eclipse in a given year, and 0 otherwise. 1(past eclipse) is a dummy variable equal to 1 for counties that have already experienced a total eclipse in previous years, and 0 otherwise. County controls include all time varying county characteristics reported in Panel A of Table 2, log agricultural land tax per capita (1820), suitability of sweet potato, latitude degrees, longitude degrees, and log distance to coast interacted with year fixed effects. Standard errors clustered at the prefecture level reported in parenthesis. Significance level: *** p<0.01, ** p<0.05, * p<0.1.

# Table A.4: Conley Adjusted SEs for Main Results

| | Panel: 1(rebellion) (1) | Qing Rebellions (2) | OLS: Xinhai Rebellions (3) | RF: Xinhai Rebellions (4) | 2SLS: Xinhai Rebellions (5) |
|---|---|---|---|---|---|
| Coefficient | 0.117 | 0.255 | 0.816 | 0.261 | 1.025 |
| Conley Standard Errors: | | | | | |
| 50 km | (0.011) | (0.041) | (0.208) | (0.107) | (0.418) |
| 100 km | (0.014) | (0.044) | (0.209) | (0.114) | (0.445) |
| 200 km | (0.020) | (0.047) | (0.212) | (0.119) | (0.467) |
| 300 km | (0.023) | (0.049) | (0.236) | (0.128) | (0.502) |

**Notes:** This table presents the standard errors after spatial adjustment using the method of Conley (2010). The four rows with distances test the results of spatial correlation at radius of 50, 100, 200, and 300 km. Column 1 corresponds to the Column 3 of Table 3 with independent variable of 1(eclipses) in panel data; Column 2 corresponds to Column 3 of Table 4 with independent variable of N(eclipses) during Qing in cross-sectional data; Columns 3-5 correspond to Column 2, 4, and 6 of Table 5, testing the impact of rebellions before 1900 on the Xinhai Revolution using OLS, reduced-form, and IV. All specifications are the most saturated specifications. County controls include log agricultural land tax per capita (1820), suitability of sweet potato, latitude degrees, longitude degrees, log distance to coast, and the number of eclipses from 2000BCE to 3000CE excluding the Qing dynasty.

# Table A.5: Half-life Test using 1902-1911 Protests

| | N Xinhai Protests per 1,000 ppl | | | | |
|---|---|---|---|---|---|
| | 1815-1899 | 1731-1814 | 1647-1730 | 1563-1646 | 1368-1646 |
| | (1) | (2) | (3) | (4) | (5) |
| N Total Eclipses | 0.377* | 0.440* | 0.100 | -0.067 | -0.109 |
| | (0.209) | (0.224) | (0.155) | (0.195) | (0.127) |
| Observations | 1550 | 1550 | 1550 | 1550 | 1550 |
| R-squared | 0.152 | 0.154 | 0.149 | 0.149 | 0.150 |
| N clusters (prefectures) | 249 | 249 | 249 | 249 | 249 |
| Counties with Eclipses | 468 | 551 | 538 | 398 | 1091 |
| Province FE | y | y | y | y | y |
| County pop categories FE | y | y | y | y | y |
| County controls | y | y | y | y | y |

**Notes:** The table shows the results of the half-life test, using the same specification as Table 5 Column 4. Each column corresponds to a different time period. The independent variables are number of total eclipses for time periods in each column. County controls include log agricultural land tax per capita (1820), suitability of sweet potato, latitude degrees, longitude degrees, log distance to coast, and the number of eclipses from 2000BCE to 3000CE excluding the Qing dynasty. Standard errors clustered at the prefecture level reported in parenthesis. Significance level: *** p<0.01, ** p<0.05, * p<0.1.

## Table A.6: The Effect of Qing Eclipses on 1902-1911 Protests controlling Ming Eclipses

| | N Xinhai Protests per 1,000 ppl | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| (N Total Eclipses)$_{1647-1899}$ | 0.261** | 0.260* | 0.266** |
| | (0.131) | (0.133) | (0.130) |
| (N Total Eclipses)$_{1563-1646}$ | | -0.041 | |
| | | (0.192) | |
| (N Total Eclipses)$_{1368-1646}$ | | | -0.121 |
| | | | (0.121) |
| Observations | 1,550 | 1,550 | 1,550 |
| R-squared | 0.155 | 0.155 | 0.155 |
| Province FE | y | y | y |
| County pop categories FE | y | y | y |
| County controls | y | y | y |
| N clusters (prefectures) | 249 | 249 | 249 |

**Notes:** The table shows reduced form estimation results controlling eclipses in different Ming period. The dependent variable is the sum of Xinhai-related protests per 1,000 people recorded between 1902 and 1911 in each county. The independent variable is the number of total eclipses observed in the county from 1647 to 1899 (baseline, col 1), from 1563 to 1646 (late Ming, col 2), and from 1368 to 1646 (the whole Ming, col 3). Variables in per 1,000 people are constructed using county population estimates from Skinner et al. (2011). County controls include log agricultural land tax per capita (1820), suitability of sweet potato, latitude degrees, longitude degrees, log distance to coast, and the number of eclipses from 2000BCE to 3000CE excluding the Qing dynasty. Standard errors clustered at the prefecture level reported in parenthesis. Significance level: *** p<0.01, ** p<0.05, * p<0.1.

# Table A.7: Past Rebellions and Socio-Economic Development

| | log(pop) 1900 (1) | government infrastructure 1647-1899 (2) | tax shortage 1647-1899 (3) |
|---|---|---|---|
| (N Rebellions per 1,000 ppl)$_{1647-1899}$ | -0.038 | -0.003 | 0.067 |
| | (0.190) | (0.011) | (0.083) |
| Observations | 719 | 1,550 | 1,550 |
| R-squared | 0.064 | 0.004 | 0.015 |
| Province FE | y | y | y |
| County pop categories FE | y | y | y |
| County controls | y | y | y |
| N clusters (prefectures) | 158 | 249 | 249 |

**Notes:** The dependent variable for column 1 is the log of county population observed in 1900 from Hu (1935) with only 719 counties available. The dependent variable for column 2 is the cumulative investment by the central government in new infrastructure located in a given county during the Qing period. The dependent variable for column 3 is the cumulative number of years in which the taxes collected in a given county were lower than the fixed quota established by the Qing court. County controls include log agricultural land tax per capita (1820), suitability of sweet potato, latitude degrees, longitude degrees, log distance to coast, and the probability of eclipses from 2000BCE to 3000CE excluding the Qing dynasty. Standard errors clustered at the prefecture level reported in brackets. Significance level: *** p<0.01, ** p<0.05, * p<0.1.

# Table A.8: Keyword Definitions and 2SLS Results

| Group | Keywords | | | | | | | | | | | | | Y = N (Rebellions per 1,000 ppl), 1647–1899 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 勇 Bravery | 斗 Fight | 烈 Martyrdom | 捐 Sacrifice | 抗 Resistance | 廉 No Corruption | 孝 Filial | 名 Good Reputation | 豪 Generous | 骨 Fallen Soldiers | 哀 Sympathy | 忠 Loyalty | 英 Heroic | X = Legend Dummy 2SLS | | X = Count of Martyrs 2SLS | | $N$ Counties |
| 1 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | | 0.138** | 0.126** | 0.274* | 0.249* | 356 |
| 2 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | 0.121** | 0.138** | 0.317* | 0.319* | 448 |
| 3 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | 0.112* | 0.129** | 0.316* | 0.316* | 456 |
| 4 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | 0.127** | 0.141** | 0.335* | 0.333* | 461 |
| 5 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 0.121** | 0.138** | 0.346* | 0.347** | 465 |
| 6 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | 0.094 | 0.120* | 0.352* | 0.337* | 548 |
| 7 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 0.105* | 0.125** | 0.398* | 0.385* | 564 |
| Province FE and County Population Category FE | | | | | | | | | | | | | | y | y | y | y | |
| County Controls | | | | | | | | | | | | | | n | y | n | y | |

**Note:** This table tests robustness of the results when using different keyword combinations to identify martyrs. County controls include log agricultural land tax per capita (1820), suitability of sweet potato, latitude degrees, longitude degrees, log distance to coast, and the probability of eclipses from 2000BCE to 3000CE excluding the Qing dynasty. Standard errors clustered at the prefecture level reported in brackets. Significance level: *** p<0.01, ** p<0.05, * p<0.1.